Empirical measures in dynamical systems

Lua Viana Reis

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"Everyone else has an attractor — why can't we have one of our own?
Stranger than all the rest. Look at it: it's the most beautiful thing on the map."

— Maria, "Unstable Orbits in the Space of Lies" (Greg Egan).

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Nomenclature

$\mathbb{Z}$	The integers.
$\mathbb{N}$	The natural numbers, usually starting at zero.
$\mathbb{Z}_{\geq 0}$	The natural numbers, definitely starting at zero.
$\mathbb{Z}_{\geq 1}$	The natural numbers, definitely starting at one.
$\mathcal{M}(X)$	The set of measures of a measurable space X, with the weak-$*$ topology.
$\mathcal{M}_1(X)$	The set of probability measures of a measurable space X, with the weak-$*$ topology.
$\mu_n \xrightarrow{w} \mu$	Weak-$*$ convergence.
$L^p(\mu)$	The space of p-norm integrable functions for μ modulo the a.e. equality relation.
$L^p(\mathcal{A})$	The subspace of $L^p(\mu)$ generated by functions that are $\mathcal{A}$-measurable.
$\mathbb{E}_\mu(\phi \mid \mathcal{A})$	Conditional expectation of ϕ with respect to the measure μ and σ-algebra $\mathcal{A}$.
$(\Omega, \mathcal{F}, \mathbb{P})$	Some arbitrary probability space.
$\mathcal{B}(\mu)$	Basin of the measure μ, defined in section 1.3.
W	The Wiener measure, defined in appendix A.4.
$C(X)$	The set of continuous functions $X \to \mathbb{R}$ with the topology of uniform convergence.
$\mathbb{D}([0,1])$	The Skorohod space, defined in appendix A.4.

Chapter 1

Ergodic theory and dynamics

1.1 A categorical zoo

The starting point of the field of dynamical systems is a space X, endowed with some structure, along with some notion of "time evolution", which could be a map $X \to X$ preserving the said structure. For instance, in topological dynamics, X is usually a topological space and the time evolution is given by a continuous map, while in ergodic theory the space is endowed with the structure of a measure and the maps are measure-preserving. But it is also of equal interest to the field to look at more versatile settings, like the ones in which there are multiple maps — or perhaps, the ones in which the maps do not send send each given state to a single next state, but to a set, or distribution, of future possibilities.

Of course, we could have a separate definition for each (and that is usually what we do). But it is also fun to see how general we can make a reasonable definition that still conveys what most such systems are. One interesting way to do that, which nicely bundles in a single notion the totality of the examples above, is the one of a *monoid object* in a category.

Following [1], we can define a monoid object in a category C as a functor $M \to C$, where M is *monoid*, that is, a category with a single object. An usual monoid is just a group without the inverse operation, namely, a set M endowed with a special identity element 1_M and a multiplication $(a, b) \in X \times X \mapsto a \cdot b \in X$ satisfying

i. $1_M \cdot a = a = a \cdot 1_M$ for all $a \in M$;

ii. $(a \cdot b) \cdot c = a \cdot (b \cdot c)$ for all $a, b, c \in M$.

Given such a monoid M, the category composed of a single object $\bigstar$, set of morphisms $\mathrm{Hom}(\bigstar, \bigstar) = M$ and composition given by the monoid multiplication will satisfy the

definition of a category. Conversely, any (small) category with a single object $\bigstar$ can be seen as a monoid, by taking the carrier set to be the set $M = \mathrm{Hom}(\bigstar, \bigstar)$ and the product, to be the composition. Let's see how many familiar examples of dynamics fit neatly in this definition.

Examples

One map

The simplest example is that of a set X and a map $f\colon X \to X$. This fits the categorical definition by taking $C = \mathbf{Set}$ and $M = \mathbb{N}$ (the natural numbers under addition, which forms a monoid). The functor $F\colon M \to \mathbf{Set}$ sends the single object $\bigstar$ of M to X, and each morphism $n \in \mathbb{N} = \mathrm{Hom}(\bigstar, \bigstar)$ to the n-fold composition $f^n = f \circ f \circ \cdots \circ f$ (with $f^0 = \mathrm{id}_X$). The functor axioms ensure that $F(0) = \mathrm{id}_X$ and $F(n+m) = F(n) \circ F(m) = f^n \circ f^m = f^{n+m}$, which is precisely the monoid homomorphism property.

Two maps

To illustrate the definition, if $f_a, f_b\colon X \to X$ are two measure-preserving maps, then the associated monoid is the one-object category for the free monoid with two generators $\{a, b\}$, and the functor sends $\bigstar$ to the object X and words to the compositions of f_a and f_b in the category of measure-preserving maps. For instance, the word $abba$ is sent to the map $f_a \circ f_b \circ f_b \circ f_a\colon X \to X$.

Markov kernels

Another example is that of Markov kernels, or Markov chains. Often, it is useful to model stochastic dynamics by having the maps send each point $x \in X$ not to a point $f(x) \in X$, but to a *probability measure* $f(x) \in \mathcal{M}_1(X)$. The probability measure $f(x)$ then encodes the distribution of expected outcomes given that we started in the state x.

How we can compose iterates of f and regard it as a dynamic on X? Semantically, we want the iterates to encode the distribution of expected outcomes after n steps given that the system started at the point x. To work it out, we can start with the case where X is finite, so any $f(x) \in \mathcal{M}_1(X)$ is completely determined by some transition probabilities $\mathbb{P}(f(x) = y) := f(x)(\{y\})$. Then, supposing that the probability of first making the transition $x \mapsto y$ and then $y \mapsto z$ is $\mathbb{P}(f(x) = y) \cdot \mathbb{P}(f(y) = z)$, we should have

$$\mathbb{P}(f^2(x) = z) = \sum_{y \in X} \mathbb{P}(f(x) = y) \cdot \mathbb{P}(f(y) = z).$$

If we rewrite the above with a measure-theoretic language, we see that we ought to define

$$f^2(x)(E) = \int_X f(y)(E)\ d(f(x))\{y\}.$$

More generally, given $f\colon X \to \mathcal{M}_1(Y)$, $g\colon Y \to \mathcal{M}_1(Z)$, we can define a kernel composition by

$$(g \circ f)(x)(E) = \int_X g(y)(E)\ d(f(x))\{y\}$$

so that $g \circ f\colon X \to \mathcal{M}_1(Z)$.

A map with type $X \to \mathcal{M}_1(Y)$ is called a *Markov kernel* between the measurable spaces X and Y. The category whose objects are measurable spaces and morphisms are Markov kernels, with compositions given by the rule above, will be called **Stoch**. We remark that the identity morphisms in this category are the Dirac masses $\delta(x) = \delta_x \in \mathcal{M}_1(X)$.

An interesting aspect of this category is that deterministic measurable maps still live inside it — given $g\colon X \to Y$ a measurable function, $f_g(x) = \delta_y$ is a Markov kernel. That is to say, this correspondence defines a (faithful) functor from the category **Meas** of measurable functions to **Stoch**. This means that dynamics in **Stoch** can be seen a generalization of deterministic measurable systems, where we now also allow the maps to be stochastic.

Another observation is that a probability measure μ in a measurable space X can be represented in this category by a Markov kernel $m\colon \{\star\} \to \mathcal{M}_1(X)$, where $\{\star\}$ is a measurable space with one element. Namely, we set $m(\star) = \mu$.

Invariant states and observables

Two important concepts in the study of dynamical systems are the notions of invariant states and invariant observables. In [1], a systematic treatment of those objects is suggested using the categorical language which we have been developing here.

A *cone* over a functor $F\colon \mathcal{D} \to \mathcal{C}$ is defined as an object $C \in \mathcal{C}$ and a natural transformation $T\colon \Delta_C \Rightarrow F$, where $\Delta_C\colon \mathcal{D} \to \mathcal{C}$ is the constant functor at the identity of C. In the case we are interested, $\mathcal{D} = M$ is a monoid and F is given by the dynamics. This means a cone is given as an object $C \in \mathcal{C}$ and a morphism $c\colon C \to X$ with the property that for every $m \in M$,

$$\begin{tikzcd} & C \arrow[dl, "c"'] \arrow[dr, "c"] & \\ X \arrow[rr, "m"] & & X \end{tikzcd}$$

commutes.

A cone (C, c) is *universal* for $M \to C$ if it is a limit of the above diagram, that is, for every other cone (D, d), there is a unique arrow $D \to C$ so that

$$\begin{tikzcd} & D \arrow[d, dashed] \arrow[dl, "d"'] \arrow[dr, "d"] & \\ & C \arrow[dl] \arrow[dr] & \\ X \arrow[rr, "m"] & & X \end{tikzcd}$$

commutes. If a cone is to be seen as a collection of invariant states, then the universal cone is the biggest of them. As we shall see in appendix B.4, a universal cone is a special case of a *limit*. Since categorical limits are unique, if a universal cone exists, it is unique up to isomorphism as well.

Example 1.1. Let $f\colon X \to X$ be any function from a set to itself. Like in previous example, this will define a dynamical system in **Set** over the monoid $\mathbb{Z}_{\geq 0}$, mapping $n \in \mathbb{Z}_{\geq 0}$ to the iterate f^n. If $x \in X$ is a fixed point of f, then we get a cone by defining $C = \{x\}$ and picking the morphism $c\colon C \to X$ to be the inclusion. Since x is a fixed point, it's clear that the associated diagram will commute. Conversely, if (C, c) is a cone and $x \in C$, then the commutativity of the triangle means $f(c(x)) = c(x)$, and therefore $c(x) \in X$ is a fixed point for f.

In this case, the *universal* cone for f, then, will correspond to the set $\mathrm{Fix}(f)$ of all fixed points (which may be empty) paired with the set inclusion $\mathrm{Fix}(f) \hookrightarrow X$. Any other cone will map uniquely to this set, by matching each of its points with the corresponding fixed point in $\mathrm{Fix}(f)$.

Example 1.2. Let $f\colon X \to \mathcal{M}_1(X)$ be a *Markov kernel* from a measurable set to itself. Examples of cones for the associated system in **Stoch** over $\mathbb{Z}_{\geq 0}$ are *stationary measures* $m\colon \{\star\} \to \mathcal{M}_1(X)$, that is, measures with the property that

$$(f \circ m)(\star) = \int_X f(x)\, d\mu\{x\} = \mu,$$

where $\mu = m(\star)$. In other words, the following diagram commutes in the category **Stoch**:

$$\begin{tikzcd} & \{\star\} \arrow[dl, "m"'] \arrow[dr, "m"] & \\ X \arrow[rr, "f"] & & X \end{tikzcd}$$

More generally, a kernel $g\colon Y \to \mathcal{M}_1(X)$ will define a cone if for every $y \in Y$, $g(y) \in \mathcal{M}_1(X)$ is invariant. As we shall see later, if X is a Borel space and f comes from a deterministic map (that is, every $f(x)$ is a Dirac mass), then the universal cone exists and is given by what is called the *ergodic decomposition*.

1.2 Ergodic theory and ergodicity

Ergodic theory is the domain of dynamical systems that study dynamics by means of invariant measures; it can be seen as the specialization of the section above to categories related to measure spaces and measure-preserving transformations. The hope is that the invariant measures are able to encode the asymptotic statistical behavior of the dynamics, even when individual points have very chaotic and unpredictable trajectories, and therefore, we should turn our attention to them. Indeed, the cornerstone theorem of the field, the Birkhoff pointwise ergodic theorem, provides a very positive answer to this hope.

The pointwise ergodic theorem

Theorem 1.3. *Let μ be an invariant probability measure and $\phi\colon X \to \mathbb{R}$ an integrable function. Then, for μ-almost every $x \in X$,*

$$\frac{1}{n}\sum_{i=0}^{n-1} \phi(f^i(x)) \to \mathbb{E}_\mu(\phi \mid \mathcal{I})(x), \tag{1.1}$$

where $\mathcal{I}$ is the sub-σ-algebra of (strictly) invariant sets.

Below we reproduce a succinct proof of the theorem, described for example in [2, p. 136][1].

Proof. Let

$$F_n(x) = \max_{1\le k\le n} \sum_{i=0}^{k} \phi(f^i(x)).$$

Claim 1. $F_{n+1} = \phi + \max\{0, F_n \circ f\}$.

To see this, note that $\sum_{i=0}^{k} \phi \circ f^i = \phi + \sum_{i=0}^{k-1} \phi \circ f^{i+1}$, so

$$F_{n+1} = \phi + \max_{1\le k\le n+1} \sum_{i=0}^{k-1} \phi \circ f^{i+1} = \phi + \max\left\{0, \max_{1\le k\le n} \sum_{i=0}^{k} \phi \circ f^{i+1}\right\} = \phi + \max\{0, F_n \circ f\}.$$

Claim 2. $F_{n+1} - F_n \circ f = \phi - \min\{0, F_n \circ f\}$.

This claim follows from the previous one by rewriting

$$F_{n+1} + \min\{0, F_n \circ f\} = \phi + \max\{0, F_n \circ f\} + \min\{0, F_n \circ f\} = \phi + F_n \circ f.$$

[1] This is not exactly relevant to this dissertation, but the author has formally verified this proof in the Lean 4 ITP. The code is currently available at `https://github.com/lua-vr/pointwise-birkhoff`. It is in the process of being upstreamed to the Mathlib library.

Two very useful corollaries of the claim are that for each $x \in X$, $F_{n+1}(x) - F_n(f(x))$ is a non-increasing sequence (because $F_n(f(x))$ is non-decreasing by definition), and that if $F_n(f(x)) \to \infty$, then the sequence will converge monotonically to $\phi(x)$.

Now, we define the set

$$S = \{x \in X \mid F_n(x) \to +\infty\}.$$

The first observation is that $S \in \mathcal{I}$, because it's measurable and strictly invariant. Since μ is invariant by f, this implies $\int_S F_n \circ f \, d\mu = \int_S F_n \, d\mu$. Also, the observations from the last claim show that $F_{n+1}(x) - F_n(f(x)) \to \phi(x)$ for $x \in S$. Thus, by the dominated convergence theorem,

$$0 \leq \int_S F_{n+1} - F_n \, d\mu = \int_S F_{n+1} - F_n \circ f \, d\mu \to \int_S \phi \, d\mu \tag{1.2}$$

which means $\int_S \phi \, d\mu \geq 0$.

The second observation is that if $x \notin S$, then $(F_n(x))_n$ is bounded above, so

$$\limsup_{n\to\infty} \frac{1}{n} \sum_{i=0}^{n-1} \phi(f^i(x)) \leq \limsup_{n\to\infty} \frac{1}{n} F_n(x) \leq 0. \tag{1.3}$$

Putting the two together, we conclude that if ϕ is such that $\mathbb{E}(\phi|\mathcal{I}) < 0$ almost everywhere, then by eq. (1.2) and the fact $S \in \mathcal{I}$, the set S has measure zero, so eq. (1.3) is valid almost everywhere.

The theorem then follows by noticing that for any $\varepsilon > 0$, if we replace ϕ by $\tilde{\phi} = \phi - \mathbb{E}(\phi \mid \mathcal{I}) - \varepsilon$, then $\mathbb{E}(\tilde{\phi} \mid \mathcal{I}) < 0$, so

$$\limsup_{n\to\infty} \frac{1}{n} \sum_{i=0}^{n-1} \tilde{\phi}(f^i(x)) = \limsup_{n\to\infty} \frac{1}{n} \sum_{i=0}^{n-1} \phi(f^i(x)) - \mathbb{E}(\phi \mid \mathcal{I}) - \varepsilon \leq 0.$$

If we combine this with the analogous inequality for $-\phi$, we get

$$\mathbb{E}(\phi \mid \mathcal{I}) - \varepsilon \leq \liminf_{n\to\infty} \frac{1}{n} \sum_{i=0}^{n-1} \phi(f^i(x)) \leq \limsup_{n\to\infty} \frac{1}{n} \sum_{i=0}^{n-1} \phi(f^i(x)) \leq \mathbb{E}(\phi \mid \mathcal{I}) + \varepsilon$$

and the result follows by letting $\varepsilon \to 0$. □

The Birkhoff ergodic theorem shows that the Birkhoff averages often look much simpler than the observable itself. Indeed, $\mathbb{E}(\phi \mid \mathcal{I})$ is measurable for the sub-σ-algebra $\mathcal{I}$, and therefore is an invariant function: $\mathbb{E}(\phi \mid \mathcal{I}) \circ f = \mathbb{E}(\phi \mid \mathcal{I})$. If this algebra is trivial, then all the Birkhoff averages would converge to the same value. This suggests the following definition.

Definition 1.4. We say that a measure μ is *ergodic* for f if the algebra $\mathcal{I}$ is trivial modulo μ: for every $A \in \mathcal{I}$, either $\mu(A) = 0$ or $\mu(A^c) = 0$.

1.3 Empirical measures

Let X be a compact metric space, and $f\colon X \to X$ be a Borel measurable map. Given a point $x \in X$, we can define a sequence of *empirical measures*

$$e_n(x) = \frac{1}{n}\sum_{i=0}^{n-1} \delta_{f^i(x)},$$

encoding the averages along the orbit of x up to a time n. These measures are, indeed, uniquely defined by the property that for any observable $\phi\colon X \to \mathbb{R}$, its integral with respect to $e_n(x)$ equals the Birkhoff average up to time n:

$$\int \phi \; d(e_n(x)) = \frac{1}{n}\sum_{i=0}^{n-1} \phi(f^i(x)).$$

The direct relation to Birkhoff averages also highlights these objects as having a central role in ergodic theory. In particular, weak convergence of the empirical measures mean that asymptotically, the orbit has a well defined statistics: there is some invariant measure μ which encode the limit of the Birkhoff averages of all continuous observables ϕ.

Lemma 1.5. *If $e_n \xrightarrow{w} \mu$, then μ is invariant for f.*

Proof. Since X is compact and μ is a Borel, continuous functions are dense in $L^1(\mu)$. Therefore, it suffices to check that $\mu(\phi) = \mu(\phi \circ f)$ for all $\phi \in C(X)$. For that, we notice that

$$|e_N(\phi) - e_N(\phi \circ f)| \leq 2 \max \phi / N,$$

and let $N \to \infty$. □

Reciprocally, for each invariant measure $\mu \in \mathcal{M}_f(X)$, we can ask what are the points of X whose empirical measures converge to it. The set $\mathcal{B}(\mu) \subseteq X$ of points with this property is called the *basin* of the measure μ.

Lemma 1.6. *If μ is an invariant probability for f, then $\mu(\mathcal{B}(\mu)) \in \{0, 1\}$. Moreover, $\mu(\mathcal{B}(\mu)) = 1$ if, and only if, μ is ergodic for f.* new :)

Proof. Suppose μ is an ergodic for f. Since X is compact, there is a sequence $\{\phi_i\}_{i\in\mathbb{N}} \subseteq C(X)$ of functions that is uniformly dense in $C(X)$. By the pointwise ergodic theorem, for each i there is a full measure set E_i where $e_n(x)(\phi_i) \to \mu(\phi_i)$ for all $x \in E_i$. Let $E = \bigcap_{i\in\mathbb{N}} E_i$, which has full measure.

We claim that for all $\phi \in C(X)$ and $x \in E$, we have $e_n(x)(\phi) \to \mu(\phi)$. Let $\varepsilon > 0$ and let i be such that $\|\phi - \phi_i\|_\infty < \varepsilon$. If N is large enough that $n \geq N$ implies $\left|e_n(x)(\phi_i) - \mu(\phi_i)\right| < \varepsilon$, we have

$$\left|e_n(x)(\phi) - \mu(\phi)\right| \leq \left|e_n(x)(\phi_i) - \mu(\phi_i)\right| + \left|e_n(x)(\phi - \phi_i)\right| + \left|\mu(\phi - \phi_i)\right| \leq 3\varepsilon.$$

This shows the claim. In particular, $E \subseteq \mathcal{B}(\mu)$, and therefore $\mu(\mathcal{B}(\mu)) = 1$.

Now, suppose μ is not ergodic; in this case, there exists some invariant set $E \subseteq X$ with $\mu(E) \in (0,1)$. For $\varepsilon > 0$, consider a continuous function ϕ with the property that $\mu(\phi) = 1$ and $\int_E \phi \, d\mu < \varepsilon$, which exists by normality and inner regularity. In one hand, for all $x \in \mathcal{B}(\mu)$, we have $e_n(x)(\phi) \to \mu(\phi)$, so

$$\int_{\mathcal{B}(\mu)\cap E} e_n(x)(\phi) \, d\mu \to \mu(\mathcal{B}(\mu) \cap E)\mu(\phi) = \mu(\mathcal{B}(\mu) \cap E).$$

On the other hand, by the pointwise ergodic theorem and the fact $\mathcal{B}(\mu) \cap E$ is invariant,

$$\int_{\mathcal{B}(\mu)\cap E} e_n(x)(\phi) \, d\mu \to \int_{\mathcal{B}(\mu)\cap E} \mathbb{E}_\mu(\phi \mid \mathcal{I})(x) \, d\mu = \int_{\mathcal{B}(\mu)\cap E} \phi(x) \, d\mu < \varepsilon.$$

Therefore, $\mu(\mathcal{B}(\mu) \cap E) < \varepsilon$. Letting $\varepsilon \to 0$, we have $\mu(\mathcal{B}(\mu) \cap E) = 0$. Exchanging E with E^c, we conclude that $\mu(\mathcal{B}(\mu)) = 0$. □

Lemma 1.7. *Let X be a compact metric space with the Borel σ-algebra. For every $x \in X$, the limit points in the weak-$*$ topology of the sequence of empirical measures*

$$\mathcal{M}(x) = \{e_n(x)\}'_{n\geq 0} = \bigcap_{i=0}^{\infty} \overline{(\{e_n(x)\}_{n\geq i})}$$

form a compact connected subset of the invariant measures.

Proof. The set is closed by definition, so by Banach-Alaoglu it's compact; for connectedness, suppose $\mathcal{M}(x) = K_1 \cup K_2$ where K_1, K_2 are closed and disjoint. Let $\{\phi_i\}_{i\in\mathbb{N}} \subseteq B_{C(X)}$ be a dense subset and define

$$d(\mu, \nu) = \sum_{i=1}^{\infty} \frac{|\mu(\phi_i) - \nu(\phi_i)|}{2^i},$$

which is a metric inducing the weak-$*$ topology on the set of probability measures (see C.3). Since it is a metric, there are two open sets $U_1 \supseteq K_1, U_2 \supseteq K_2$ with positive distance $s > 0$ from each other. We claim $\lim_{n\to\infty} d(e_n(x), e_{n+1}(x)) = 0$, which would contradict $s > 0$ because the definition of $\mathcal{M}(x)$ implies $e_n(x) \in U_1$ and $e_{n+1}(x) \in U_2$ for infinitely many n. Indeed,

$$\begin{aligned} d(e_n(x), e_{n+1}(x)) &= \sum_{i=1}^{\infty} \frac{|\frac{1}{n}\sum_{i=0}^{n-1} \phi_i \circ f^i(x) - \frac{1}{n+1}\sum_{i=0}^{n} \phi_i \circ f^i(x)|}{2^i} \\ &\leq \sum_{i=1}^{\infty} \frac{2}{n} \frac{\|\phi_i\|_\infty}{2^i} \leq \frac{2}{n} \to 0. \end{aligned}$$

□

1.4 Palis' dream conjecture

Describe the overall problem of generically finding finite set of limiting measures, whose basins cover almost all points of the manifold. Describe the robust counterexample to finiteness by Newhouse. Describe that the problem of historical behavior is open.

Theorem 1.8 (Newhouse [3]). *Let M be a compact surface. There exists a non-empty open set $U \subseteq \mathrm{Diff}^2(M)$ of non-hyperbolic diffeomorphisms. Moreover, any diffeomorphism in a dense G_δ-subset of U has infinitely many sinks.*

We note that in this context, a *sink* is an attracting periodic orbit. Each sink corresponds to an ergodic measure (supported on the period orbit) whose basin is an open set, and therefore has positive volume.

Conjecture 1. *A*

1.5 Relating dynamics: extensions

Different dynamical systems can be related to each other by the concept of *factors*, or *extensions*, expressing a sense in which one system is a simplification of another. In the categorical language we have been developing, an extension is a natural transformation $F \Rightarrow G$ between two dynamical systems F and G. In this situation, we then say that G is a *factor* of F. The single morphism $F(\star) \to G(\star)$ specified by the natural transformation is also called the *factor map*.

Definition 1.9. Let $F, G\colon M \to \mathcal{C}$ be two dynamical systems on the category $\mathcal{C}$. An extension $\eta\colon F \Rightarrow G$ consists of a morphism $\eta\colon F(\star) \to G(\star)$ of $\mathcal{C}$ (the factor map), such that for all $m \in M$, the diagram

$$\begin{tikzcd} F(\star) \arrow[r, "F(m)"] \arrow[d, "\eta"] & F(\star) \arrow[d, "\eta"] \\ G(\star) \arrow[r, "G(m)"] & G(\star) \end{tikzcd}$$

commutes.

Now, we conclude our tour around the zoo and focus on our beloved species, measure-preserving systems.

Definition 1.10. Let $(X, \mathcal{B}_X, \mu, f)$ and $(Y, \mathcal{B}_Y, \nu, g)$ be measure-preserving transformations. $(Y, \mathcal{B}_Y, \nu, g)$ is a **factor** of $(X, \mathcal{B}_X, \mu, f)$ if there exists a measure-preserving map

$\phi : X \to Y$ such that

$$\begin{tikzcd} X \arrow[r, "f"] \arrow[d, "\phi"] & X \arrow[d, "\phi"] \\ Y \arrow[r, "g"] & Y \end{tikzcd}$$

commutes.

Example 1.11. Every system admits a trivial factor:

$$\begin{tikzcd} X \arrow[r, "f"] \arrow[d, "\phi"] & X \arrow[d, "\phi"] \\ \{\star\} \arrow[r, "\mathrm{id}"] & \{\star\} \end{tikzcd}$$

Lemma 1.12. *Suppose $p\colon (X, f) \to (Y, g)$ is a factor map. Then, if $\mu \in \mathcal{M}_f(X)$ is ergodic for f then $p_*\mu \in \mathcal{M}_g(Y)$ is ergodic for g.*

Proof. Suppose $A \subseteq Y$ is measurable and invariant for g. Then, $p^{-1}(A)$ is an invariant set for f, because

$$f^{-1}(p^{-1}(A)) = p^{-1}(g^{-1}(A)) = p^{-1}(A).$$

Therefore, $p_*\mu(A) = \mu(p^{-1}(A)) = 0$. □

Invertible extensions

$$\begin{tikzcd} (\widetilde{X}, \tilde{f}) \arrow[dd, "\pi"'] & \\ & (Y, g) \arrow[ul, dashed, "\exists! h"'] \arrow[dl, "p"] \\ (X, f) & \end{tikzcd} \tag{1.4}$$

Theorem 1.13. *Let $(X, \mathcal{B}, \mu, f)$ be a measure-preserving system on a Borel space. Then, the natural invertible extension exists.*

Proof. Apply Kolmogorov extension theorem.

□

Lemma 1.14. *Let $\pi\colon (\widetilde{X}, \tilde{f}, \tilde{\mu}) \to (X, f, \mu)$ be the natural invertible extension. Then, we have a converse for lemma 1.12: if μ is ergodic for f, then $\tilde{\mu}$ is ergodic for $\tilde{f}$.*

Proof. Suppose μ is ergodic but $\tilde{\mu}$ is not; then, we can write $\tilde{\mu} = \lambda\mu_0 + (1-\lambda)\mu_1$ for $\mu_0 \neq \mu_1$. Let $\nu_0 = \pi_*\mu_0$ and $\nu_1 = \pi_*\mu_1$, so that $\mu = \lambda\nu_0 + (1-\lambda)\nu_1$ by linearity. By ergodicity, $\nu_0 = \nu_1 = \mu$, contradicting the uniqueness of the measure $\tilde{\mu}$ such that $\pi_*\tilde{\mu} = \mu$. □

Invertible extensions may not exist for non-Borel spaces. The following example of non-existence of the natural extension expands on an exercise of [4], and relies on the Axiom of Choice.

Example 1.15 (Non-existence of the natural extension)**.** Let $X = \mathbb{S}^1$ be the circle with the Borel σ-algebra $\mathcal{B}$ and $f\colon \mathbb{S}^1 \to \mathbb{S}^1$ be an irrational rotation. With the Axiom of Choice, we can form a set E with exactly one point from each orbit of f. Since f is a Lebesgue measure-preserving bijection, $E_n = f^n(E)$ are disjoint, have the same outer measure and $\bigcup_{n\in\mathbb{Z}} E_n = \mathbb{S}^1$. In particular, the E_n cannot be measurable, thus $m^*(E) > 0$.

If K is a subset of positive outer measure and $\mathcal{E} = \sigma(\{K\} \cup \mathcal{B})$, and if we define $\mu^*\colon 2^X \to [0,1]$ as

$$\mu^*(A) = \frac{m^*(A \cap K)}{m^*(K)},$$

then μ^* is an probability outer measure that restricts itself to a probability measure μ on $\mathcal{E}$. It is also the case that $\mathcal{E} = \mathcal{B} \cup \{A \cap K : A \in \mathcal{B}\} \cup \{A \cap K^c : A \in \mathcal{B}\}$. Let C_K be the completion by μ^* of the σ-algebra $\mathcal{E}$.

Let's apply the above construction for $K = \bigcup_{n=1}^{\infty} f^n(E)$. Note that $E \cap K = \varnothing$, therefore $\mu^*(E) = 0$ and $E \in C_K$. We conclude that $f^{-1}(K) = E \cup K$ is measurable as well, so for every $A \in \mathcal{B}$, $f^{-1}(A \cap K) = f^{-1}(A) \cap f^{-1}(K)$ is measurable, thus $f^{-1}(\mathcal{E}) \subseteq C_K$.

Now, for any $A \in C_K$, we can write $A = B \cup Z$ with $B \in \mathcal{E}$ and $\mu^*(Z) = 0$; thus, $f^{-1}(A) = f^{-1}(B) \cup f^{-1}(Z)$. Notice that

$$\mu^*(f^{-1}(Z)) = m^*(Z \cap f(K)) \leq m^*(Z \cap K) = \mu^*(Z) = 0,$$

therefore $f^{-1}(C_K) \subset C_K$.

The same calculation shows that $\mu^*(f^{-1}(A)) \leq \mu^*(A)$ for any $A \subseteq X$. Thus, if A is measurable,

$$1 - \mu(f^{-1}(A)) = \mu(f^{-1}(A^c)) \leq \mu(A^c) = 1 - \mu(A),$$

so $f_*\mu = \mu$ on C_K.

Consider $\Sigma = \{(x_i)_{i\leq 0} : f(x_i) = x_{i+1}\} \subseteq X^{\mathbb{Z}_{\leq 0}}$ as the inverse limit of (X, f, C_K), endowed with the restriction of the product σ-algebra. Note that $\tilde{f}\colon \Sigma \to \Sigma$ given by

$$\tilde{f}((x_i)_{i\leq 0}) = (f(x_i))_{i\leq 0} = (\ldots, x_{-1}, x_0, f(x_0))$$

is invertible and measurable, since

$$\tilde{f}([A_{-k}, \ldots, A_0] \cap \Sigma) = [A_{-k}, \ldots, A_0, X] \cap \Sigma.$$

For each $n \in \mathbb{Z}_{\leq 0}$, let $K_n = [n \mid K] = \{(x_i)_{i\leq 0} : x_n \in K\}$. Suppose there exists $\tilde{\mu}$, a measure on Σ such that $\pi_*\tilde{\mu} = \mu$, where $\pi((x_i)_{i\leq 0}) = x_0$. Then

$$\tilde{\mu}(K_n) = \tilde{\mu}(\tilde{f}^n(\pi^{-1}(K))) = \mu(K) = 1$$

for every n. On the other hand, since f is invertible,

$$\begin{aligned}\bigcap_{n\leq 0} K_n &= \{(x_i)_{i\leq 0} : \forall i, x_i \in K\} \\ &= \{(x_i)_{i\leq 0} : \forall n \geq 0, f^{-n}(x_0) \in K\}.\end{aligned}$$

But this set is empty because

$$\{(x_i)_{i\leq 0} : \forall n \geq 0, f^{-n}(x_0) \in K\} = \bigcap_{n=0}^{\infty} f^n(K) = \limsup_n E_n = \varnothing,$$

a contradiction.

Question 1.1. Is there an example that does not use the Axiom of Choice?

The solenoid

[5] [6]

1.6 Cocycles and skew products

$$\begin{tikzcd} X \times Y \arrow[r] \arrow[d, "\pi_X"] & X \times Y \arrow[d, "\pi_X"] \\ X \arrow[r, "f"] & X \end{tikzcd}$$

Example 1.16 (Iterated function system)**.** IFSs are examples of skew products.

Lyapunov exponents

Define for linear cocycles. Oseledets theorem.

1.7 Partitions

A *partition* of a set S is a collection of disjoint subsets $\mathcal{P} \subseteq 2^S$ whose union is S. It's important to notice that $\mathcal{P}$ can be seen either as a set of subsets or (with some abuse of notation) as a function $\mathcal{P} \colon S \to 2^S$ assigning to each point $p \in S$ the unique $\mathcal{P}(p) \in \mathcal{P}$ such that $p \in \mathcal{P}(p)$. To make matters clearer, henceforward we shall denote by $\overline{\mathcal{P}} \subseteq 2^S$ the image of $\mathcal{P} \colon S \to 2^S$ whenever we need to emphasize that the set of subsets is the object of our attentions.

Lattice structure

The partitions $\mathbf{P}(S)$ of a set S form a lattice under the refinement order. Namely, we endow $\mathbf{P}(S)$ with the partial order where $\mathcal{P} \preceq \mathcal{Q}$ if and only if for all $Q \in \overline{\mathcal{Q}}$, there is some $P \in \overline{\mathcal{P}}$ such that $Q \subseteq P$, and whenever this happens we say that $\mathcal{Q}$ is *finer* and $\mathcal{P}$ is *coarser* relative to each other. Given two partitions $\mathcal{P}, \mathcal{Q} \in \mathbf{P}(S)$, there is a *join* operation defined by

$$(\mathcal{P} \vee \mathcal{Q})(x) = \mathcal{P}(x) \cap \mathcal{Q}(x)$$

that is the least upper bound of $\mathcal{P}$ and $\mathcal{Q}$ (the coarsest partition finer than both $\mathcal{P}$ and $\mathcal{Q}$). There is also a *meet* operation $\mathcal{P} \wedge \mathcal{Q}$, giving the greatest lower bound, which is less straightforward to construct: on the union $\overline{\mathcal{P}} \cup \overline{\mathcal{Q}}$, we say P and Q are related if they overlap, that is, $P \cap Q \neq \varnothing$, and by taking the transitive closure of this relation we get an equivalence relation on the set $\overline{\mathcal{P}} \cup \overline{\mathcal{Q}}$. The elements of $\mathcal{P} \wedge \mathcal{Q}$, then, are precisely the unions of sets in the same equivalence class:

$$\mathcal{P} \wedge \mathcal{Q} = \left\{ \bigcup_{S \in [P]} S \,\middle|\, [P] \in \overline{\mathcal{P}} \cup \overline{\mathcal{Q}}/\sim \right\}.$$

The partitions $\mathbf{P}(S)$ form, in fact, a *complete* lattice, meaning that all subsets $\mathsf{P} \subseteq \mathbf{P}(S)$ have both a least upper bound $\bigvee_{\mathcal{P} \in \mathsf{P}} \mathcal{P} \in \mathbf{P}(S)$ and a greatest lower bound $\bigwedge_{\mathcal{P} \in \mathsf{P}} \mathcal{P} \in \mathbf{P}(S)$. Their construction is very similar to their binary versions: the join can be defined as $\left(\bigvee_{\mathcal{P} \in \mathsf{P}} \mathcal{P}\right)(x) = \bigcap_{\mathcal{P} \in \mathsf{P}} \mathcal{P}(x)$, and we can use the join to define the meet as

$$\bigwedge_{\mathcal{P} \in \mathsf{P}} \mathcal{P} = \bigvee_{\substack{\mathcal{Q} \text{ is an upper} \\ \text{bound for } \mathsf{P}}} \mathcal{Q}.$$

In the case of $\mathsf{P} = \{\mathcal{P}, \mathcal{Q}\}$, they reduce to the binary operations of the paragraph above.

An interlude of zero measure

In a measure space, it is often useful to relax the notions of equality and order on the partitions so that zero measure sets are disregarded. This is done in the obvious way, namely by saying $\mathcal{P} \preceq \mathcal{Q}$ if for all $Q \in \overline{\mathcal{Q}}$, there is a $P \in \overline{\mathcal{P}}$ such that $Q \setminus P$ has zero measure. Going a bit further, the partitions themselves may be allowed to cover only a total measure subset of the space, and their elements could overlap on zero measure sets and still be regarded as disjoint. Formally, the idea here is to quotient the subsets by the almost-surely equality relation, and notice that the usual binary operators on the sets lift to well-defined operators on the quotient.

1.8 Disintegrations

In the Calculus course, we learn that it's most often easier to find the area of a region in the plane if we reduce the problem back to a single dimension, by taking integrals of the length along parallel lines. This is the essence of Fubini's theorem, which states that the integral of a function in a higher dimensional region can be written as nested integrals of the function along one-dimensional subsets. Then we go to the probability course, and the lecturer talks about conditional expectations and the whole philosophy of Bayes of taking probabilities conditioned to some event. It may not seem at first that this has anything to do with the aforementioned theorem in the Calculus course, at least not until we take the Ergodic theory course and learn about measures and disintegrations.

While Fubini remains true for product measures, in general the problem of decomposing an arbitrary measure into a family of measures defined in smaller subsets is not as straightforward. Neither is straightforward to see how could one talk about the probability of an event conditioned to another event of *zero probability*, even though in practice this happens very often. Both situations are instances of the general problem the disintegration of measures solves: given a measure μ and a partition $\mathcal{P}$ of the space X, we want to find a family of measures $\{\mu_x\}_{x\in X}$, each μ_x giving full measure to $\mathcal{P}(x)$, with the property that $\mu(E) = \int \mu_x(E)\, d\mu(x)$.

Definition 1.17 (Disintegration)**.** A *disintegration* of a measure $\mu \in \mathcal{M}(X)$ with respect to a partition $\mathcal{P}$ is a family $\{\mu_x\}_{x\in X} \subseteq \mathcal{M}(X)$ such that for every measurable set $E \subseteq X$,

1. $\mu_x(\mathcal{P}(x)) = 1$ for μ-almost every x;
2. the map $x \mapsto \mu_x(E)$ is measurable;
3. $\mu(E) = \int \mu_x(E)\, d\mu(x)$.

Uniqueness

Measurability

Not all partitions are created equal. Some may be pathological enough that there may be no hope in disintegrating over them. In fact, such partitions appear *very commonly* in dynamics. A prime example is the partition consisting of orbits of a map without too many periodic points.

Example 1.18. Suppose $f\colon X \to X$ is invertible and μ is an invariant measure for f such that $\mu(X \setminus \mathrm{Per}(f)) > 0$. Then, the partition $\mathcal{O}_f\colon X \to 2^X$ of two-sided orbits of f is such that μ does not admit a disintegration with respect to $\mathcal{O}_f$.

Proof. Suppose that such disintegration $\{\mu_x\}_{x\in X}$ exists. We claim that $\{f_*\mu_x\}_{x\in X}$ is also a disintegration. Indeed, all $O_f(x)$ are invariant, so we have

$$(f_*\mu_x)(O_f(x)) = \mu_x(f^{-1}(O_f(x))) = \mu_x(O_f(x)) = 1$$

for a.e. x, and similarly, we have that $x \mapsto (f_*\mu_x)(E) = \mu_x(f^{-1}(E))$ is measurable, and $\mu(E) = \mu(f^{-1}(E)) = \int f_*\mu_x(E)\, d\mu(x)$. By uniqueness, this implies that $f_*\mu_x = \mu_x$ almost everywhere.

By item 1, the hypothesis and the paragraph above, we can find some $x \notin \mathrm{Per}(f)$ such that $f_*\mu_x = \mu_x$ and $\mu_x(O_f(x)) = 1$. But this is impossible: since $O_f(x)$ is infinite and $\mu_x(\{f^k(x)\}) = \mu_x(\{x\})$ for every k,

$$1 = \mu_x(O_f(x)) = \sum_{k\in\mathbb{Z}} \mu_x(\{f^k(x)\}) = \sum_{k\in\mathbb{Z}} \mu_x(\{x\}) \in \{0, \infty\}.$$

□

Remark 1. A very basic instance of example 1.18 would be with $X = S^1$, $f\colon S^1 \to S^1$ an irrational rotation, and μ the unique ergodic measure, because f has no periodic points.

Remark 2. The example 1.18 also shows that it is not sufficient for the *elements* of the partition to be measurable sets. In the remark above, the orbits are countable sets of points in the circle, and thus are Lebesgue (or even Borel) measurable, and yet the partition does not admit a disintegration.

The right notion of a "well-behaved" partition with respect to disintegrations will turn out to be the notion of *measurable partitions*.

Definition 1.19. A partition $\mathcal{P}$ is said to be *measurable* if there is a sequence $\mathcal{P}_1, \mathcal{P}_2, \ldots$ of partitions, all of which composed of a countable number of measurable sets, such that $\mathcal{P} = \bigvee_{n\in\mathbb{N}} \mathcal{P}_n$.

Example 1.20. The partition $\perp$ of all points of a second-countable T_1 topological space X is measurable in the Borel σ-algebra.

Proof. Let $\{U_n\}_{n\in\mathbb{N}}$ be a countable basis for the topology of X, and define $\mathcal{P}_n = \{U_n, U_n^c\}$. Let $P \in \mathcal{P} = \bigvee_{n\in\mathbb{N}} \mathcal{P}_n$, which is measurable by definition, and $p \in P$. If $q \in P$, then we must have $\mathcal{P}_n(p) = \mathcal{P}_n(q)$ for all $n \in \mathbb{N}$, or in other words, $p \in U_n \iff q \in U_n$, for all n. The T_1 property then implies $p = q$, so $P = \{p\}$. This shows $\mathcal{P} = \perp$. □

Example 1.21. Any partition $\mathcal{P}$ of a compact metric space X whose elements are closed is measurable in the Borel σ-algebra.

Proof. Since they are closed, the elements of $\mathcal{P}$ are measurable in the Borel algebra. To show the partition is countably generated, we can define a metric on $\overline{\mathcal{P}}$ by letting $d(P_1, P_2)$ be the distance between the compact sets $P_1, P_2 \in \overline{\mathcal{P}}$ (this is the Hausdorff metric, which we detail in **??**). With this metric, the surjection $\mathcal{P} \colon X \to \overline{\mathcal{P}}$ is a contraction, so $\overline{\mathcal{P}}$ is a compact metric space as well. By the example 1.20, the partition $\perp$ of points of $\overline{\mathcal{P}}$ is countably generated, thus so is its preimage by $\mathcal{P}$. □

Conditional measures

1.9 Entropy

Topological entropy

Define.

Measure-theoretic entropy

Define.

Pesin entropy formula

State the theorem, to be used in the invariance principle section.

Chapter 2

Schwarzian derivatives and cylinder maps

In this chapter, we follow the exposition by Bonifant and Milnor [7] on a family of endomorphisms of a cylinder with interesting properties.

2.1 Cross ratios and the Schwarzian derivative

Suppose we start with a field $\mathbb{K}$ and, not being happy with the fact that we cannot divide by zero, decide to add a point at infinity; this gives us the *projective line* $P^1(\mathbb{K}) = \mathbb{K} \sqcup \{\infty\}$, whose arithmetic operations are the same as those of $\mathbb{K}$ except for the additional rules $\frac{1}{0} = \infty$, $\frac{1}{\infty} = 0$, $x \cdot \infty = \infty$ if $x \neq 0$, and $x + \infty = \infty$ if $x \neq \infty$. It's not possible to define such operations everywhere without introducing inconsistent rules, so the operations are partial — we cannot multiply $\infty \cdot 0$ or add $\infty + \infty$.

Our new space $P^1(\mathbb{K})$ has an interesting property: given $z \in P^1(\mathbb{K})$, we have three types of operations that we can apply to z: translations $z \mapsto z + x$, where $x \in \mathbb{K}$, homotheties $z \mapsto xz$ for $x \in \mathbb{K} \setminus \{0\}$, and the inverse $z \mapsto 1/z$, which was not defined before if $z = 0$, but now makes sense for every z. These three families of actions, which are invertible, generate the *group of homographies* with coefficients in $\mathbb{K}$.

A second way to construct $P^1(\mathbb{K})$, which makes the role of the group of homographies as a symmetry group of this space even clearer, is to think of the space of "formal fractions", that is, pairs of numbers (x, y), where $x, y \in \mathbb{K}$ are not both zero, where we identify "equivalent fractions", in the sense that $(\lambda x, \lambda y) \sim (x, y)$ for every $\lambda \in \mathbb{K} \setminus \{0\}$. The quotient by this equivalence relation gives us $P^1(\mathbb{K})$ as the space of lines passing through the origin of the vector space $\mathbb{K}^2$, and the operations are like fraction operations: $(x, y) + (z, w) = (wx + yz, yw)$, $(x, y) \cdot (z, w) = (xz, yw)$, and $(x, y)^{-1} = (y, x)$. Despite being seemingly more complicated, the interesting fact about this construction is that it

emphasizes how the symmetry group of the space $\mathbb{K}^2$, the linear group $\mathrm{GL}_2(\mathbb{K})$, naturally induces a symmetry group on the quotient, the projective group $\mathrm{PGL}_2(\mathbb{K})$, which is precisely the group of homographies.

This group is a quotient of $\mathrm{GL}_2(\mathbb{K})$ by a subgroup of dimension 1, so it has dimension 3. Furthermore, it is 3-transitive: given two sets of three distinct points in $P^1(\mathbb{K})$, there exists an element $g \in \mathrm{PGL}_2(\mathbb{K})$ that takes the first set to the second. But not only that, $P^1(\mathbb{K})$ has three special points: 0, 1, and ∞. Thus, given $x_1, x_2, x_3 \in P^1(\mathbb{K})$ distinct, we can ask which $g \in \mathrm{PGL}_2(\mathbb{K})$ such that $g(x_1) = \infty$, $g(x_2) = 0$, and $g(x_3) = 1$. With some calculations, we arrive at

$$g(z) = \frac{(x_2 - z)(x_3 - x_1)}{(x_1 - z)(x_3 - x_2)}$$

The *cross ratio* of four points x_0, x_1, x_2, x_3 is the value of $g(x_0)$ for the element g of the homography group that sends $x_1 \mapsto \infty$, $x_2 \mapsto 0$, and $x_3 \mapsto 1$.

Definition 2.1 (Cross ratio)**.**

$$\rho(x_0, x_1, x_2, x_3) = \frac{(x_2 - x_0)(x_3 - x_1)}{(x_1 - x_0)(x_3 - x_2)}$$

Definition 2.2. Let $f\colon I \to I'$ be a C^3 diffeomorphism between two intervals of the real line. The *Schwarzian derivative* of f is the function

$$\mathcal{S}f = \frac{f'''}{f'} - \frac{3}{2}\left(\frac{f''}{f'}\right)^2.$$

We shall write $\mathcal{S}f \prec 0$ if there exists a dense open set $U \subseteq I$ such that $\mathcal{S}f(y) < 0$ for every $y \in U$. Similarly, we write $\mathcal{S}f \succ 0$ if there exists $U \subseteq I$ open and dense such that $\mathcal{S}f(y) > 0$ for every $y \in U$.

The Schwarzian derivative has some important properties that we highlight in the Proposition below, as described in [7, Proposition B.1].

Proposition 2.3. *The Schwarzian derivative has the following properties:*

1. *If $\mathcal{S}f \succ 0$ and $\mathcal{S}g \succ 0$, then $\mathcal{S}(f \circ g) \succ 0$, and similarly for $\prec 0$. Moreover, if $\mathcal{S}f \equiv 0$, then $\mathcal{S}(f \circ g) = \mathcal{S}g$, and if $\mathcal{S}g \equiv 0$, then $\mathcal{S}(f \circ g) = (\mathcal{S}f) \circ g$.*
2. *If $\mathcal{S}f \prec 0$, then $\mathcal{S}f^{-1} \succ 0$.*
3. *$\mathcal{S}f \prec 0$ if, and only if, $\varphi(y) = 1/\sqrt{|f'(y)|}$ is strictly convex.*
4. *$\mathcal{S}f \equiv 0$ if, and only if, f is a fractional linear transformation.*

Proof.

1. We have
$$\mathcal{S}(f\circ g) = \frac{(f\circ g)'''}{(f\circ g)'} - \frac{3}{2}\left(\frac{(f\circ g)''}{(f\circ g)'}\right)^2.$$
By the Faà di Bruno's (beautiful) formula [8, p. 479] in its combinatorial form,
$$(f\circ g)^{(n)}(x) = \sum_{\pi\in\Pi} f^{(|\pi|)}(g(x))\cdot\prod_{B\in\pi} g^{(|B|)}(x)$$
where the sum is taken over the set Π of all partitions of $\{1,\dots,n\}$, and $B\in\pi$ means B is an element of the partition π (therefore, a subset of $\{1,\dots,n\}$).
Specializing Faà di Bruno's formula to $n = 1, 2, 3$, we have
$$\begin{aligned}(f\circ g)' &= (f'\circ g)\cdot g'\\ (f\circ g)'' &= (f''\circ g)\cdot (g')^2 + (f'\circ g)\cdot g''\\ (f\circ g)''' &= (f'''\circ g)\cdot (g')^3 + 3(f''\circ g)\cdot g''\cdot g' + (f'\circ g)\cdot g'''\end{aligned}$$
and thus
$$\begin{aligned}\mathcal{S}(f\circ g) &= \frac{(f'''\circ g)\cdot (g')^3 + 3(f''\circ g)\cdot g''\cdot g' + (f'\circ g)\cdot g'''}{(f'\circ g)\cdot g'}\\ &\qquad - \frac{3}{2}\left(\frac{(f''\circ g)\cdot (g')^2 + (f'\circ g)\cdot g''}{(f'\circ g)\cdot g'}\right)^2\\ &= \frac{(f'''\circ g)\cdot (g')^2}{f'\circ g} + 3\frac{(f''\circ g)\cdot g''}{f'\circ g} + \frac{g'''}{g'} - \frac{3}{2}\left(\frac{(f''\circ g)\cdot g'}{f'\circ g} + \frac{g''}{g'}\right)^2\\ &= \left(\frac{f'''\circ g}{f'\circ g} - \frac{3}{2}\left(\frac{f''\circ g}{f'\circ g}\right)^2\right)\cdot (g')^2 + \frac{g'''}{g'} - \frac{3}{2}\left(\frac{g''}{g'}\right)^2\\ &= (\mathcal{S}f)\circ g\cdot (g')^2 + \mathcal{S}g. \qquad (2.1)\end{aligned}$$
Since g is a diffeomorphism, $g'(x)\neq 0$ for all $x\in I$, and therefore $(g')^2 > 0$. In particular, g is a homeomorphism, so that if $\mathcal{S}f(x) > 0$ for all x the open dense set U, and $\mathcal{S}g(x) > 0$ in the open dense set V, then $\mathcal{S}(f\circ g) > 0$ in the open dense set $g^{-1}(U)\cap V$. Similarly, from eq. (2.1) we see that if $\mathcal{S}f\equiv 0$, then the first term is zero, therefore $\mathcal{S}(f\circ g) = \mathcal{S}g$.

2. Applying eq. (2.1) to $\mathrm{id} = f^{-1}\circ f$, we see that
$$0 = \mathcal{S}(\mathrm{id}) = (\mathcal{S}f^{-1})\circ f\cdot (f')^2 + \mathcal{S}f,$$
and by the same argument of the paragraph above, we conclude $\mathcal{S}f < 0 \iff \mathcal{S}f^{-1} > 0$.

3. Since f is a diffeomorphism between intervals, either $f' > 0$ or $f' < 0$. As a first step, assume $f' > 0$. Taking the second derivative of $\varphi(y) = 1/\sqrt{f'(y)} = f'(y)^{-1/2}$,
$$\varphi'' = \left(-\frac{1}{2}(f')^{-3/2}\cdot f''\right)' = \frac{3}{4}(f')^{-5/2}\cdot (f'')^2 - \frac{1}{2}(f')^{-3/2}\cdot f'''.$$

Thus

$$2\varphi''/\varphi = \frac{3}{2}(f')^{-4/2} \cdot (f'')^2 - (f')^{-2/2} \cdot f''' = \frac{3}{2}\left(\frac{f''}{f'}\right)^2 - \frac{f'''}{f'} = -\mathcal{S}f. \tag{2.2}$$

From this, and noticing that $\varphi > 0$, we see that if $\mathcal{S}f(x) < 0$ if and only if $\varphi''(x) > 0$. A C^2 function $\varphi : I \to \mathbb{R}$ will be strictly convex if and only if $\varphi'' \geq 0$ and φ'' does not vanish on any open interval, so $\mathcal{S}f \prec 0$ implies that ϕ'' is positive in an open dense set, thus strictly convex. Reciprocally, if φ is strictly convex, then the set $\{x \mid \varphi''(x) > 0\} = \{x \mid \mathcal{S}f(x) < 0\}$ is open and its complement does not contain any open interval, and therefore is dense.

If $f' < 0$, we define $g\colon I \to -I'$, $g(y) = -f(y)$, so that $g' > 0$ and we fall back to the previous case. The argument is finished by noticing that $\mathcal{S}g = \mathcal{S}f$ and $1/\sqrt{|f'|} = 1/\sqrt{|g'|}$.

4. From eq. (2.2), we see that $\mathcal{S}f \equiv 0 \iff \varphi'' \equiv 0$. This, in turn, is equivalent to φ being linear, say $\varphi(y) = cy + d$. We thus have

$$f'(y) = \frac{1}{(cy+d)^2}.$$

Integrating both sides, by the Fundamental Theorem of Calculus we have

$$f(y) - f(a) = \int_a^y \frac{1}{(ct+d)^2}\, dt = \frac{1}{c}\left(\frac{1}{ca+d} - \frac{1}{cy+d}\right) = \frac{y-a}{c(ca+d)y + d(ca+d)}$$

which is a fractional linear transformation, as desired.

□

Lemma 2.4. *If $f\colon I \to I'$ is C^3 and $\mathcal{S}f \prec 0$, then f has at most three fixed points. When it has three fixed points, the middle one is strictly repelling and the other two are strictly attracting.*

Proof. Suppose f had four fixed points $y_1 < y_2 < y_3 < y_4$. By the Mean Value Theorem, in each of the three open intervals (y_i, y_{i+1}), for $i = 1, 2, 3$, we would have a point $c_i \in (y_i, y_{i+1})$ where $f'(c_i) = 1$. But this is impossible, since by item 3 $\varphi(y) = 1/\sqrt{|f'(y)|}$ is strictly convex, which would imply $\varphi(c_2) < \varphi(c_1) = \varphi(c_3) = 1$.

Now, suppose there are exactly three fixed points $y_1 < y_2 < y_3$. By the same reasoning, there are two points $c_1 \in (y_1, y_2)$ and $c_2 \in (y_2, y_3)$ where $\varphi(c_1) = \varphi(c_2) = 1$. Knowing that φ is strictly convex, this implies $\varphi(y_2) > 1$ and $\varphi(y_1), \varphi(y_3) < 1$, and therefore $|f'(y_2)| < 1$ and $|f'(y_1)|, |f'(y_3)| > 1$. □

Lemma 2.5. *If $f\colon I \to I'$ is C^3, then $\mathcal{S}f \prec 0$ if, and only if, f increases cross ratios, in the sense that $\rho(f(y_1), f(y_2), f(y_3), f(y_4)) > \rho(y_1, y_2, y_3, y_4)$ for all $y_1 < y_2 < y_3 < y_4$.*

Proof. Given $y_1 < y_2 < y_3 < y_4$, let φ be the linear fractional transformation that sends $f(y_1)$ to y_1, $f(y_2)$ to y_2 and $f(y_4)$ to y_4. Since from proposition 2.3, we have that $\mathcal{S}(\varphi \circ f) = \mathcal{S}f$, and we know fractional linear transformations also preserve cross-ratios, without loss of generality we may assume that f itself fixes y_1, y_2 and y_4. Now, by lemma 2.4, $f(y) > y$ on the interval (y_2, y_4), and therefore $f(y_3) > y_3$. It thus follows easily by the definition of cross-ratios that $\rho(f(y_1), f(y_2), f(y_3), f(y_4)) > \rho(y_1, y_2, y_3, y_4)$. □

Lemma 2.6. *If $f\colon I \to I$ is a diffeomorphism that preserves orientation and $\mathcal{S}f \prec 0$, where $I = [a, b]$, then $f'(a)f'(b) < 1$.*

Proof. Let's consider the case where $f'(a) = f'(b)$ first. If we had $f'(a)^2 = f'(b)^2 = f'(a)f'(b) \geq 1$, then $\varphi(a) = \varphi(b) \leq 1$ which by strict convexity of φ implies $\varphi|_{(a,b)} < 1$. This, in turn, means that $f'|_{(a,b)} < 1$, which (by the mean value theorem) is not possible for a diffeomorphism that sends an interval to itself.

The general case is shown by noticing that if we define the function $g = r \circ f \circ r\colon I \to I$, where $r(x) = a + b - x$ (and thus g is the reflection of f along the anti-diagonal), then $g'(a) = f'(b)$ and $g'(b) = f'(a)$. By the chain rule,

$$(f \circ g)'(a) = (f \circ g)'(b) = f'(a)f'(b),$$

and by item 1, $\mathcal{S}(f \circ g) \prec 0$ because $\mathcal{S}(r) \equiv 0$ and thus $\mathcal{S}(g) = \mathcal{S}(f) \circ r \prec 0$. □

Note that by proposition 2.3, analogous results to the last two lemmas hold if $\mathcal{S}f \succ 0$. For example, if $\mathcal{S}f \succ 0$ then f decreases cross ratios and $f'(0)f'(1) > 1$.

2.2 Skew products

We call a map $F\colon X \times Y \to X \times Y$ a *skew product* if it is of the form $F(x, y) = (g(x), f_x(y))$ for some $g\colon X \to X$ and a family of functions $\{f_x\}_{x \in X}$, $f_x\colon Y \to Y$. We are interested in the case where X is a compact metric space with the Borel σ-algebra, $g\colon X \to X$ is a continuous map preserving an ergodic probability measure $\mu \in \mathcal{M}(X)$, $Y = I$ is the interval $[0, 1]$, and $f_x\colon I \to I$ are C^3 diffeomorphisms preserving orientation. We will see how, depending on the Schwarzian derivative of the maps f_x, we observe different behaviors of the dynamics with respect to the invariant measures $\mu_0 = \mu \otimes \delta_0$ and $\mu_1 = \mu \otimes \delta_1$. Note that for $i \in \{0, 1\}$, $\operatorname{supp} \mu_i = \mathcal{A}_i$, where $\mathcal{A}_i = X \times \{i\}$ are the upper and lower "boundaries" of $X \times I$.

To simplify notation, let's denote by $f_x^n(y)$ the second coordinate of $F^n(x, y)$, that is, $f_x^n = f_{g^{n-1}(x)} \circ \cdots \circ f_x$.

Lemma 2.7. *If $(x, y_1) \in \mathcal{B}(\mu_0)$, then $(x, y_2) \in \mathcal{B}(\mu_0)$ for all $y_2 \leq y_1$. Similarly, if $(x, y_1) \in \mathcal{B}(\mu_1)$, then $(x, y_2) \in \mathcal{B}(\mu_1)$ for all $y_2 \geq y_1$.*

Proof. We will prove the case $i = 0$, as the situation is symmetric. Let $(x, y_1) \in \mathcal{B}(\mu_0)$ and $y_2 \leq y_1$. If $\varphi \in C(X \times I)$, and $\varepsilon > 0$, by the uniform continuity of φ, there exists $\delta > 0$ such that $y < \delta$ implies $|\varphi(x, 0) - \varphi(x, y)| < \varepsilon$ for all $x \in X$. Since $U = X \times [0, \delta)$ is a continuity set for μ_0 and $e_n(x, y_2) \longrightarrow \mu_0$, by Portmanteau

$$\lim_{n\to\infty} e_n(x, y_2)(U) = \mu_0(U) = 1.$$

Define $\tau_n = e_n(x, y_2)(U) = \frac{1}{n}\#\{0 \leq i < n \mid f_x^n(y_2) < \delta\}$. Note that if $f_x^n(y_2) < \delta$, then $f_x^n(y_1) < \delta$ since the diffeomorphisms preserve orientation. Thus,

$$\begin{aligned}|e_n(x, y_1)(\varphi) - e_n(x, y_2)(\varphi)| &\leq \frac{1}{n}\sum_{i=0}^{n-1}\left|\varphi(g^n(x), f_x^n(y_1)) - \varphi(g^n(x), f_x^n(y_2))\right| \\ &\leq \varepsilon\tau_n + 2\|\varphi\|_\infty(1 - \tau_n).\end{aligned}$$

Since $\tau_n \longrightarrow 1$, there exists N such that $n \geq N$ implies that the right side is less than 2ε, hence both integrals on the left converge to $\mu_0(\varphi)$. □

2.3 Transversal Lyapunov exponents

To each point (x, y) of either boundary $\mathcal{A}_i$, for $i \in \{0, 1\}$, we can associate a number that expresses the average expansion rate of tangent vectors along the direction normal to the skew product boundary. This is a specific case of the general definition of Lyapunov exponents, whose existence is given by the Oseledets theorem. make the link to section 1.6

Definition 2.8 (Transversal Lyapunov exponent)**.**

$$\mathrm{Lyap}_{\mathcal{A}_i}(x) = \lim_{n\to\infty}\frac{1}{n}\log\left|\partial_y F^n(x, i)\right|.$$

By the product rule, and noting that $\{0, 1\}$ are fixed points of the fiber diffeomorphisms, the right side of the equation above can be rewritten as

$$\lim_{n\to\infty}\frac{1}{n}\sum_{k=0}^{n-1}\log\left|\partial_y F(g^k(x), i)\right| = \lim_{n\to\infty}\frac{1}{n}\sum_{k=0}^{n-1}\log f'_{g^k(x)}(i).$$

Thus, by the ergodicity of μ for g and the Birkhoff ergodic theorem, for μ-almost every $x \in X$ it holds that

$$\mathrm{Lyap}_{\mathcal{A}_i}(x) = \int \log f'_x(i)\, d\mu(x), \tag{2.3}$$

and we shall denote this common value by $\mathrm{Lyap}(\mathcal{A}_i)$.

The statement below assumes less regularity than the corresponding one in [7, lemma A.1], that asks for F to be C^2. Also, the second part of the proof (when the Lyapunov

exponent is positive) is original, because the basins of the original article are topological instead of measure-theoretic. In our context, $F(x, y) = (g(x), f_x(y))$ where g is an ergodic map for μ and $f_x \in C^1[0, 1]$ for all x. In particular, if $(x, y) \mapsto f_x'(y)$ is continuous, the following equicontinuity hypothesis is satisfied.

Lemma 2.9. *If the family $\{f_x'\}_{x\in X}$ is equicontinuous at $i \in \{0, 1\}$ and $\mathrm{Lyap}(\mathcal{A}_i) < 0$, then the basin of $\mu \otimes \delta_i$ has positive $\mu \otimes \mathrm{Leb}$ measure. Conversely, if $\mathrm{Lyap}(\mathcal{A}_i) > 0$, then the corresponding basin has null measure.*

Proof. We will assume $i = 0$. Let $\varepsilon > 0$ be such that

$$\int \log(f_x'(0) + \varepsilon)\, d\mu(x) < 0.$$

Let $a(x) = \log(f_x'(0) + \varepsilon)$. By the Birkhoff ergodic theorem, for μ-almost every $x \in X$

$$\lim_{n\to\infty} \frac{1}{n} \sum_{k=0}^{n-1} a(g^k(x)) = \int a\, d\mu < 0,$$

which implies $\lim_{n\to\infty} \sum_{k=0}^{n-1} a(g^k(x)) = -\infty$. This means that the function

$$A(x) = \max_{n\geq 1} \sum_{k=0}^{n-1} a(g^k(x)) \geq 0$$

is well-defined in almost every x.

By equicontinuity, there is $\delta > 0$ such that $0 \leq y \leq \delta$ implies $f_x'(x) \leq f_x'(0) + \varepsilon$ for all $x \in X$. Thus, if $y < \delta e^{-A(x)}$, we claim that

$$f_x^n(y) \leq \exp\left(\sum_{k=0}^{n-1} a(g^k(x)) - A(x)\right)\delta \leq \delta. \tag{2.4}$$

is true by induction. Indeed, the $n = 0$ case follows from the hypothesis on y, and if (2.4) is true for n, then by the mean value theorem,

$$\begin{aligned} f_x^{n+1}(y) &\leq \left(\max f_{g^n(x)}'([0, f_x^n(y)])\right) f_x^n(y) \\ &\leq (f_{g^n(x)}'(0) + \varepsilon) f_x^n(y) \\ &\leq \exp\left(a(g^n(x)) + \sum_{k=0}^{n-1} a(g^k(x)) - A(x)\right)\delta. \end{aligned}$$

Since $\lim_{n\to\infty} \sum_{k=0}^{n-1} a(g^k(x)) = -\infty$, we have $f_x^{n+1}(y) \to 0$. Thus, if $x \in \mathcal{B}(\mu)$ (which has full measure), then $e_n(x, y) \xrightarrow{w} \mu_0$ whenever $y \leq \delta e^{-A(x)}$.

The converse is slightly more tricky, because (x, y) may be in the basin of μ_0 but $f_x^n(y) \not\to 0$. Let $\varepsilon > 0$ be such that new :)

$$\int \log(f_x'(0) - \varepsilon)\, d\mu(x) > 0.$$

and let $M = \max_{(x,y)} |\log(f_x'(y))|$. Define $a(x) = \log(f_x'(0) - \varepsilon)$, so like before,

$$\lim_{n\to\infty} \frac{1}{n} \sum_{i=0}^{n-1} a(g^i(x)) = \int a\, d\mu > 0,$$

for $x \in E$ in a full measure set, and let δ be such that $0 \le y \le \delta$ implies $|f_x'(0) - f_x'(y)| < \epsilon$. Let $\gamma = (\int a\, d\mu)/(8M) < 1$. Take $(x, y) \in \mathcal{B}(\mu_0)$ such that $x \in E$ and $y \in (0, 1)$, which is possible if we assume $\mathcal{B}(\mu_0)$ has positive $\mu \otimes \mathrm{Leb}$ measure. Now, consider the strip $S_\delta = X \times [0, \delta)$. Since $\mu_0(S_\delta) = 1$, $\mu_0(\partial S_\delta) = 0$ and $(x, y) \in \mathcal{B}(\mu_0)$, by proposition A.6 there is N_1 for which $n \ge N_1$ implies $G_n = \{0 \le i \le n \mid f_x^i(y) < \delta\}$ satisfies $\#G_n/n > 1 - \gamma$. Let $N_2 \ge N_1$ be such that $n \ge N_2$ implies $\frac{1}{n}\sum_{i=0}^{n-1} a(g^i(x)) > (\int a\, d\mu)/2$. Then, for $n \ge N_2$ and any $z \le y$, we have

$$\frac{1}{n}\log((f_x^n)'(z)) = \frac{1}{n}\sum_{i=0}^{n-1} \log(f'_{g^i(x)}(f_x^i(z))) \ge \frac{1}{n}\sum_{i\in G_n} \log(f'_{g^i(x)}(f_x^i(z))) - \gamma M.$$

Now, $i \in G_n$ implies $f_x^i(y) < \delta$, which implies $f_x^i(z) < \delta$ by monotonicity. This, in turn, implies $\log(f'_{g^i(x)}(f_x^i(z))) \ge \log(f'_{g^i(x)}(0) - \varepsilon) = a(g^i(x))$. Therefore

$$\begin{aligned}\frac{1}{n}\log((f_x^n)'(z)) \ge \frac{1}{n}\sum_{i\in G_n} a(g^i(x)) - \gamma M. &\ge \frac{1}{n}\sum_{i=0}^{n-1} a(g^i(x)) - 2\gamma M \\ &\ge \Big(\int a\, d\mu\Big)/4 > 0.\end{aligned}$$

This means that $(f_x^n)'(z) \to \infty$ uniformly on $z \le y$ (because neither N_2 nor $\int a\, d\mu$ depend on $z \in [0, y]$). This is impossible, because by the mean value theorem, $f_x^n(y) = (f_x^n)'(z_n)\cdot y$ for some $0 \le z_n \le y$, while $f_x^n(y) \le 1$ is bounded. □

Corollary 2.10. *If $(x, y) \mapsto \mathcal{S}f_x(y)$ is continuous and $\mathcal{S}f_x < 0$ for μ-almost every $x \in X$, then* $\mathrm{Lyap}(\mathcal{A}_0) + \mathrm{Lyap}(\mathcal{A}_1) < 0$. *In particular, at least one of the basins has positive measure.*

Proof. If suffices to integrate the logarithm of the inequality $f_x'(0)f_x'(1) < 1$ from lemma 2.6, and then apply lemma 2.9. □

2.4 Intermingled basins

The basins of the examples we saw before are continuity sets for the volume measure — that is, they are open sets modulo a zero volume set — and the union of all basins have

full Lebesgue measure. But should basins of attractors constitute well-behaved sets in general? In some cases, such as for uniformly hyperbolic diffeomorphisms, this indeed the case, and there is a finite number of physical measures whose basins are continuity sets, and their union has full volume [9]. Outside the realm uniform hyperbolicity, though, this may not be the case — Kan [10] gave the first example of a partially hyperbolic endomorphism on a surface with two physical measures whose basins are intermingled, in the sense that for every open set U, the intersection of this set with either basin has positive volume [9]. This implies both basins are dense in the surface, and therefore not continuity sets.

Definition 2.11. Let X be a compact metric space, $f\colon X \to X$ and μ be a reference Borel measure. We say (X, f, μ) displays intermingled basins if there exists two distinct invariant measures $\mu_0 \neq \mu_1$ such that for each open set $U \subseteq X$, we have $\mu(\mathcal{B}(\mu_0) \cap U), \mu(\mathcal{B}(\mu_1) \cap U) > 0$.

Refer to discussion in [11]

For this section, let X be a compact manifold, $g\colon X \to X$ be ergodic for $\mu \in \mathcal{M}_1(X)$ and $F\colon X \times I \to X \times I$ a continuous skew-product endomorphism of the form $F(x, y) = (g(x), f_x(y))$. We shall understand by "volume" the reference measure $m = \mu \otimes \text{Leb}$. As mentioned in lemma 2.9, under some mild hypothesis for F, if the transverse Lyapunov exponents at the invariant boundaries $\mathcal{A}_i = X \times \{i\}$, given by

$$\text{Lyap}(\mathcal{A}_i) = \int \log f_x'(i)\, d\mu(x)$$

for $i = 0, 1$ are both negative, then both basins have positive volume. We will state this condition as an hypothesis:

Hypothesis 1. The basins $\mathcal{B}(\mu_0)$ and $\mathcal{B}(\mu_1)$ have positive volume $\mu \otimes \text{Leb}$.

Assuming this, the main property that will produce the intermingled phenomenon is the following additional hypothesis.

Hypothesis 2. The map g is topologically exact, in the sense that for each nonempty open set U, there exists n for which $g^n(U) = X$. Moreover, there exists two expanding periodic points x^+ and x^- in X, of periods n^+, n^-, such that for every $y \in (0, 1)$, $f_{x^+}^{n^+}(y) > y$ and $f_{x^-}^{n^-}(y) < y$.

Let's define the measures $\beta_i(A) = m(A \cap \mathcal{B}(\mu_i))$. Then the basins are intermingled if, and only if, $\operatorname{supp} \beta_i = X \times I$ for both $i = 0, 1$. Indeed, both supports being equal to the whole space means precisely that for every open set $U \subseteq X \times I$, the measures $\beta_i(U) = m(U \cap \mathcal{B}(\mu_i))$ are both positive for the reference measure $m = \mu \otimes \text{Leb}$.

We will show this is the case for $\operatorname{supp}\beta_0$, as the proof for $i = 1$ follows by symmetry. First, we shall collect some lemmas.

Lemma 2.12. *If $(x, y_1) \in \operatorname{supp}\beta_0$, then $(x, y_2) \in \operatorname{supp}\beta_0$ for all $y_2 \leq y_1$.*

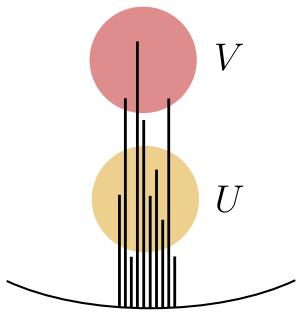
Proof. By lemma 2.7, we know this property is true for $\mathcal{B}(\mu_0)$. Let $U = B_\varepsilon(x, y_2)$ be a ball centered at (x, y_2) of radius smaller than $1 - y_1$, and $V = B_\varepsilon(x, y_1)$. Notice that

$$V \cap \mathcal{B}(\mu_0) \subseteq \{(u, v + y_1 - y_2) \mid (u, v) \in U \cap \mathcal{B}(\mu_0)\}.$$

Since $(x, y_1) \in \operatorname{supp}\beta_0$, and $m = \mu \otimes \mathrm{Leb}$ is invariant by vertical translations,

$$\beta_0(U) = m(U \cap \mathcal{B}(\mu_0)) \geq m(V \cap \mathcal{B}(\mu_0)) > 0.$$

Since this is true for all sufficiently small balls U, this shows $(x, y_2) \in \operatorname{supp}\beta_0$. □



Lemma 2.13. *The set $\operatorname{supp}\beta_0$ is backwards invariant, that is, $F^{-1}(\operatorname{supp}\beta_0) = \operatorname{supp}\beta_0$.*

Proof. If $p \in F^{-1}(\operatorname{supp}\beta_0)$ and U is a neighborhood of p, then since F is an open map, $F(U)$ is a neighborhood of $F(p) \in \operatorname{supp}\beta_0$, so $\beta_0(F(U)) > 0$. Since F is Lipschitz, if $\beta_0(U)$ were 0, then $\beta_0(F(U)) = m(F(U) \cap \mathcal{B}(\mu_0)) = m(F(U \cap \mathcal{B}(\mu_0))) = 0$, which is not the case, so $\beta_0(U) > 0$. □

Lemma 2.14. *We have $m(\operatorname{supp}\beta_0) \geq \beta_0(\operatorname{supp}\beta_0) = \beta_0(X \times I) = m(\mathcal{B}(\mu_0)) > 0$.*

Proof. The last value is positive by hypothesis 1, and the first and last equalities are, respectively, by subadditivity and definition. To show the second equality, let $C = (\operatorname{supp}\beta_0)^c$. By definition, for each $x \in C$, there exists an open neighborhood $U_x \ni x$ for which $\beta_0(U_x) = 0$. Since $X \times I$ is Lindelöf, we can extract a countable subcover $U_{x_1} \cup U_{x_2} \cup \cdots$ of C. By subadditivity, this shows $\mu(C) = 0$. □

Lemma 2.15. *The set $B_0 = \{x \in X \mid \exists y \in (0, 1),\ (x, y) \in \operatorname{supp}\beta_0\}$ is nonempty.*

Proof. It suffices to show it has positive measure. For that, let $0 < \lambda < m(\operatorname{supp} \beta_0)$, and notice that

$$m(\operatorname{supp} \beta_0 \setminus X \times [0, \lambda]) \geq m(\operatorname{supp} \beta_0) - \lambda > 0.$$

Set $E_\lambda = \operatorname{supp} \beta_0 \setminus X \times [0, \lambda]$. Then $\mu(\pi_X(E_\lambda)) \geq m(E_\lambda) > 0$, and clearly $\pi_X(E_\lambda) \subseteq B_0$. □

Corollary 2.16. *We have* $\operatorname{supp} \beta_0 = X \times I$.

Proof. Let $x_0 \in B_0$ and $y_0 \in (0, 1)$ such that $(x_0, y_0) \in \operatorname{supp} \beta_0$. Inductively, we can construct a backward orbit $\cdots \mapsto x_{-2} \mapsto x_{-1} \mapsto x_0$ of g with the property that $x_{-n} \to \mathcal{O}(x^-)$. Indeed, since x^- is an expanding periodic point, there is a neighborhood U and a partial inverse $u \colon U \to g^{-1}(U)$ with $g \circ u = \operatorname{id}|_U$ for which $u^n(x) \to \mathcal{O}(x^-)$ for all $x \in U$. And by exactness, there is some N for which $x_0 \in g^N(U)$.

Since $x_{-n} \to \mathcal{O}(x^-)$, by hypothesis 2, for every $y \in (0, 1)$ there exists n_0 such that $n \geq n_0$ implies $f_{x_{-n}}^{-n}(y_0) > y$. In particular,

$$y_{-n} = f_{x_{-n}}^{-1} \circ \cdots \circ f_{x_{-1}}^{-1}(y_0)$$

is such that $y_{-n} \to 1$. Therefore, $(x_{-n}, y_{-n}) \to (x^-, 1)$. Since $\operatorname{supp} \beta_0$ is closed and $(x_{-n}, y_{-n}) \in F^{-n}(\operatorname{supp} \beta_0)$, we conclude that $(x^-, 1) \in \operatorname{supp} \mu_0$. Now, by exactness, the set $E = \bigcup_{n=0}^{\infty} g^{-1}(\{x^-\})$ is dense in X, and 1 is a fixed point of the f_x, thus $E \times \{1\} \subseteq \operatorname{supp} \beta_0$. Taking the closure, we have $\mathcal{A}_1 \subseteq \operatorname{supp} \beta_0$, and by lemma 2.12, it is the whole $X \times I$. □

Remark 3. With Pesin theory, it is possible to show that intermingled basins are present in the more general setting of g being a hyperbolic automorphism. See the examples 3.2.1 and 3.2.2 of [9]. The argument goes as follows: pick a point $p = (x, y)$ with $y \in (0, 1)$, and consider a disk D_p transversal to the center direction (the y direction). todo

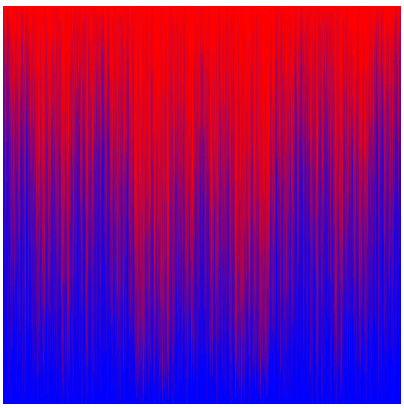


Figure 2.1: Picture of the basins from example 2.17.

Example 2.17. Define $q_a(y) = y + ay(1-y)$ and $p(x) = \varepsilon\cos(2\pi x)$ for some $0 < \varepsilon < 1$. If $|a| < 1$, then q_a is a diffeomorphism of the interval I. If $F(x) = (3x, q_{p(x)}(y))$, then $x^+ = 0$ and $x^- = 1/2$ are fixed points of the map $g(x) = 3x$ (where we identify S^1 with $[0,1]$) that satisfy hypothesis 2, because $q_a(y) > y$ if $a > 0$ and $q_a(y) < y$ if $a < 0$ for all $y \in (0,1)$. Moreover, since q_a is a degree two polynomial, it has negative Schwarzian whenever $a \neq 0$, and from the symmetry of p we see both basins of $F(x,y) = (kx, q_{p(x)}(y))$ must have positive measure, because at least one of them has. So F displays intermingled basins.

2.5 Negative Schwarzian

The statement of the theorem below corresponds to [7, Theorem 3.2], but without the hypothesis that both basins have positive measure. The question of whether this hypothesis could be removed was left open in the article, and our proof below confirms that it is unnecessary.

Theorem 2.18. *If $(x,y) \mapsto \mathcal{S}f_x(y)$ is continuous and $\mathcal{S}f_x \prec 0$ almost everywhere in $x \in X$, then there exists a measurable function $\sigma\colon X \to I$ defined almost everywhere such that*

$$(x,y) \in \mathcal{B}_0 \text{ whenever } y < \sigma(x) \quad \text{and} \quad (x,y) \in \mathcal{B}_1 \text{ whenever } y > \sigma(x).$$

Consequently, the union $\mathcal{B}_0 \cup \mathcal{B}_1$ has total $\mu \otimes \mathrm{Leb}$ measure.

Proof. We start by defining functions $\xi_0, \xi_1\colon X \to I$ that separate the basins in the following sense:

$$\xi_0(x) = \sup\{y \in I : (x,y) \in \mathcal{B}_0\}, \qquad \xi_1(x) = \inf\{y \in I : (x,y) \in \mathcal{B}_1\}.$$

By lemma 2.7, if $y < \xi_0(x)$ or $y > \xi_1(x)$, the empirical measures $e_n(x,y)$ converge to $\mu_0 = \mu \otimes \delta_0$ or to $\mu_1 = \mu \otimes \delta_1$, and if $\xi_0(x) < y < \xi_1(x)$, they do not converge to either μ_0 or μ_1.

It is sufficient to show that $A = \{x \in X : \xi_0(x) \neq \xi_1(x)\}$ has measure zero, since in this case we can define σ as the common value of the two functions. Suppose by contradiction that this set does not have measure zero. Since the f_x are bijective, A is an invariant set under g, so the ergodicity of μ implies that A has full measure.

Since we are under the hypotheses of corollary 2.10, we can assume without loss of generality that $\mathrm{Leb}(\mathcal{A}_1) > 0$. By ergodicity, we also have $\xi_1(x) < 1$ for μ-almost every $x \in X$. Thus, given $k \in \mathbb{N}$, there exists $\delta_k < 1$ such that $E_k = \{x \in X : \xi_1(x) < \delta_k\}$ has measure $\mu(E_k) > 1 - 1/k$. Applying the Birkhoff ergodic theorem to the functions 1_{E_k}, we obtain $B \subseteq X$ of full measure such that for all $x \in B$ and k,

$$\lim_{n\to\infty} \frac{1}{n}\#\{0 \le i < n \mid \xi_1(g^i(x)) < \delta_k\} = \mu(E_k) > 1 - 1/k. \tag{2.5}$$

By the ergodicity of μ, there exists another set $C \subseteq X$ of full measure whose empirical measures converge to μ, and since $\mathcal{S}f_x \prec 0$ almost everywhere in $x \in X$, there exists a set D of full measure such that $\mathcal{S}f_{g^n(x)} \prec 0$ for all $n \geq 0, x \in D$. Let $x_0 \in A \cap B \cap C \cap D$ and $y_0 \in (\xi_0(x), \xi_1(x))$, and let $\nu \in \mathcal{M}(x, y) \setminus \{\mu_0\}$ be a limit point of the empirical measures different from μ_0. Note that since $x_0 \in B$, necessarily $(\pi_X)_*\nu = \mu$, but since $\nu \neq \mu_0$, it follows that $\nu(X \times \{0\}) < 1$. Thus, $\operatorname{supp} \nu$ is not contained in $X \times \{0\}$, and there exists $\varepsilon > 0$ such that $U = X \times (\varepsilon, 1]$ satisfies $\nu(U) > 0$. Let k be such that $\nu(U) > 1/k$, and $n_j \to \infty$ such that $\liminf_{j\to\infty} e_{n_j}(x_0, y_0)(U) > 1/k$. In other words, we have

$$\liminf_{j\to\infty} \frac{1}{n_j} \#\{0 \leq i < n_j \mid f^i_{x_0}(y_0) > \varepsilon\} > 1/k. \tag{2.6}$$

Now, consider the sequence $r_n = \rho(0, f^n_{x_0}(y_0), f^n_{x_0}(\xi_1(x_0)), 1)$, which is well-defined and positive since $0 < f^n_{x_0}(y_0) < f^n_{x_0}(\xi_1(x_0)) < 1$ by the definition of the points. Note also that the graph of ξ_1 is strictly invariant under F, so $f^n_{x_0}(\xi_1(x_0)) = \xi_1(g^n(x_0))$.

Since $x \in D$, it follows by lemma 2.5 that r_n is an increasing sequence. On one hand, it is not possible for $r_n \to \infty$, because from eqs. (2.5) and (2.6) we know that there exist $n \in \mathbb{N}$ arbitrarily large such that $0 < \varepsilon < f^n_{x_0}(y_0) < f^n_{x_0}(\xi_1(x_0)) < \delta_k < 1$. On the other hand, if $r_n \to r < \infty$, then for each $x \in X$ there exists a unique $\xi(x)$ such that $\rho(0, \xi(x), \xi_1(x), 1) = r$. In fact, a calculation shows that

$$\xi(x) = \frac{\xi_1(x)}{r(1 - \xi_1(x)) + \xi_1(x)}.$$

Note that $r_n \to r$ implies $\lim_{n\to\infty} |\xi(g^n(x_0)) - f^n_{x_0}(y_0)| = 0$. Since we know that x_0 is in the basin of μ, necessarily $e_n(x_0, y_0) \xrightarrow{w} \eta$, where $\eta = (\operatorname{graph} \xi)_*\mu$ and $(\operatorname{graph} \xi)(x) = (x, \xi(x))$. Thus, η is an invariant measure. But then,

$$\varphi(x, y) = \frac{\rho(0, y, \xi_1(x), 1)}{1 + \rho(0, y, \xi_1(x), 1)}$$

is an integrable function defined at η-almost every point, with the property that $\varphi(F(x, y)) > \varphi(x, y)$ at η-almost every point. This is impossible, as it would imply $\int \varphi \, d\eta < \int \varphi \circ F \, d\eta = \int \varphi \, d\eta$. □

Example 2.19. In example 2.17, we have $q'_a(y) = 1 + a(1 - 2y)$, $q''_a(y) = -2a$ and $q'''_a \equiv 0$, so $\mathcal{S}q_a(y) < 0$ whenever $a \neq 0$. Since $q(x) = 0$ only at two points, by theorem 2.18 there exists $\sigma \colon S^1 \to I$ that separates the two basins.

2.6 Positive Schwarzian

If $f \colon M \to M$ is a diffeomorphism on a manifold, we say that an invariant measure μ is *asymptotic* with respect to a reference measure if the empirical measures of almost

every point of the reference measure converges to μ. Intuitively, μ describes the entire statistical behavior of the system, up to a set of zero volume.

Since theorem 2.18 does not require that both basins have positive measure, we also obtain a more general statement than the version in [7, Theorem 5.1], which requires both Lyapunov exponents to be positive. The proof below is also somewhat different, more similar to the proof of theorem 2.18.

Theorem 2.20. *If $\mathcal{S}f_x > 0$ almost everywhere, then F has a unique asymptotic measure ν for $\mu \otimes \mathrm{Leb}$-almost every point. Furthermore, for each $i \in \{0,1\}$, $\mathrm{Lyap}(\mathcal{A}_i) > 0$ implies $\nu(\mathcal{A}_i) = 0$.*

The idea of the proof is to pass to the invertible extension of the system and observe that the inverses of the fiber maps have negative Schwarzian. By applying theorem 2.18, we obtain an invariant graph that separates the two basins of the inverse transformation. Since the orbits diverge from the graph for the inverse of the extension, they converge to the graph for the extension. Thus, the graph describes an asymptotic measure for the extension. The pushforward of this measure is the asymptotic measure for the original map.

Proof. Let $(\tilde{X}, \tilde{g}, \tilde{\mu})$ be the invertible extension of (X, g, μ), $\pi\colon (\tilde{X}, \tilde{g}, \tilde{\mu}) \to (X, g, \mu)$ the associated conjugation, and $\tilde{F}\colon \tilde{X} \times I \to \tilde{X} \times I$ given by

$$\tilde{F}(\tilde{x}, y) = (\tilde{g}(\tilde{x}), f_{\pi(\tilde{x})}(y)).$$

Note that $\tilde{F}^{-1}(x, y) = (\tilde{g}^{-1}(x), f^{-1}_{\pi(\tilde{g}^{-1}(x))}(y))$. Since $\pi_*\tilde{\mu} = \mu$ and $\tilde{g}_*\tilde{\mu} = \tilde{\mu}$, by proposition 2.3 we have $f^{-1}_{\pi(\tilde{g}^{-1}(\tilde{x}))} < 0$ for almost every $\tilde{x} \in \tilde{X}$. Also, by lemma 1.14, $\tilde{\mu}$ is ergodic for F (and therefore, also for F^{-1}). Thus, we can apply theorem 2.18 to F^{-1} and obtain a function $\sigma\colon \tilde{X} \to I$ that separates the basins $\tilde{\mathcal{B}}_1$ and $\tilde{\mathcal{B}}_0$.

The hyperbolic metric (which coincides with the Hilbert metric) on $(0, 1)$ is given by

$$d_H(r, s) = \log|\rho(0, r, s, 1)|.$$

This metric induces the usual topology of the interval. Since maps with positive Schwarzian decrease cross-ratios, the condition $\mathcal{S}f_x > 0$ means that f_x is a contraction (not necessarily uniform) for this metric.

Let $A \subseteq \tilde{X}$ be a set of full $\tilde{\mu}$ measure where $\mathcal{S}f_{\pi(\tilde{g}^n(\tilde{x}))} > 0$ for all $n \geq 0, \tilde{x} \in A$, and $B \subseteq \tilde{X}$ be another set of full measure where the empirical measures of $\tilde{x} \in B$ converge to the ergodic measure $\tilde{\mu}$. It follows that for all $\tilde{x} \in A \cap B$ and $y \in (0, 1)$, the limit $r = \lim_{n\to\infty} d_H(f^n_{\pi(\tilde{x})}(y), \sigma(\tilde{g}^n(\tilde{x})))$ exists and is zero if and only if the orbit of $(\tilde{x}, y)$ converges in distance to the graph of σ under $\tilde{F}$, which in turn implies that $e_n(\tilde{x}, y) \xrightarrow{w} \tilde{\nu} = (\mathrm{graph}\,\sigma)_*\tilde{\mu}$.

Suppose $r > 0$. Then, as in the previous proof, there would exist another graph ξ, different from σ almost everywhere, such that $e_n(\tilde{x}, y) \xrightarrow{w} \eta = (\operatorname{graph} \xi)_* \tilde{\mu}$. Since $\varphi(\tilde{x}, y) = d_H(y, \sigma(\tilde{x}))$ is such that $\varphi(F(\tilde{x}, y)) < \varphi(\tilde{x}, y)$ whenever $(\tilde{x}, y) \in \operatorname{graph} \xi$ and η is an invariant measure for F with $\eta(\operatorname{graph} \xi) = 1$, we reach a contradiction.

This shows that $E = (A \cap B) \times (0, 1)$ is a set of full $\tilde{\mu} \otimes \mathrm{Leb}$ measure such that $(\tilde{x}, y) \in E$ implies $e_n(\tilde{x}, y) \xrightarrow{w} \tilde{\nu}$. Let $\nu = \pi_* \tilde{\nu} \in \mathcal{M}(X \times I)$. The measure ν is invariant, since

$$F_* \nu = (F \circ \pi)_* \tilde{\nu} = (\pi \circ \tilde{F})_* \tilde{\nu} = \pi_* \tilde{\nu} = \nu.$$

Furthermore, the basin of ν has full Lebesgue measure. In fact, if $\varphi \in C(X \times I)$, then $\varphi \circ \pi \in C(\tilde{X} \times I)$ and $e_n(\pi(\tilde{x}), y)(\varphi) = e_n(\tilde{x}, y)(\varphi \circ \pi)$. Therefore, $e_n(x, y) \to \nu$ if and only if $e_n(\tilde{x}, y) \to \tilde{\nu}$ for any $\tilde{x} \in \pi^{-1}(x)$, that is, $(\pi \times \mathrm{id})^{-1}(\mathcal{B}(\nu)) = \mathcal{B}(\tilde{\nu})$. It follows that $(\mu \otimes \mathrm{Leb})(\mathcal{B}(\nu)) = (\tilde{\mu} \otimes \mathrm{Leb})(\mathcal{B}(\tilde{\nu})) = 1$. □

Chapter 3

Historical behavior

Ergodic theory tries to find order in chaos by showing that even if a dynamical system looks complicated, the statistical properties of its orbits are often well behaved. But what if the statistics don't even exist for most orbits? When this happens for all orbits starting in a set of positive reference measure, the phenomenon is known as *non-statistical*, or *historical* behavior.

More precisely, by existence of statistics, we mean the weak convergence of the empirical measures $e_n(x) \xrightarrow{w} \mu$ to an asymptotic measure μ. To say that $(e_n(x))_n$ does not converge, therefore, is equivalent (in the compact setting) to the existence of a continuous observable $\varphi\colon X \to \mathbb{R}$ such that

$$\lim_{n\to\infty} \frac{1}{n} \sum_{i=0}^{n-1} \varphi(f^i(x)) \tag{3.1}$$

does not exist. (The fact that we don't need to quantify over measures that are possible limits here is a consequence of corollary C.2.) We then say that a diffeomorphism f of a compact manifold presents historical behavior if $(e_n(x))_n$ does not converge for x in a set of positive Lebesgue measure.

3.1 History

The "historical" terminology was introduced by Ruelle in [12], suggesting that for eq. (3.1) not to converge, the sequence must keep "changing its mind" as n goes to infinity, and therefore has some sort of history. For instance, in the absence of external forces like human activity, we could expect that general patterns in climate would follow some predictable statistics, while with climate change we see that such statistics are not consistent along history. In the article, Ruelle recognized the problem of *nonrecurrent* behavior of orbits as going against the expectation that recurrent behavior is generic in the

space of diffeomorphisms, and presented the open problem of finding whether historical behavior can exist in a persistent manner, making some physical justifications for this possibility. He cites the theory of spin glasses, where the time evolution of the system is akin to a random walk in a random potential: a particle stays arbitrarily large amounts of time exploring some valley in the potential before eventually crossing the barrier to another valley, in such a way that time averages like eq. (3.1) do not exist. As we shall see in this chapter, we can indeed produce historical behavior by constructing a diffeomorphism whose orbits behave like random walks, albeit not necessarily persistently.

Later, Floris Takens exposed the problem again in [13], giving some promising examples that were known at the time and also making some remarks about generiticity of points without asymptotic measures. For this reason and his work in related matters, the open question also became known as *Taken's Last Problem*.

Example 3.1 (Logistic family)**.** The family of interval maps $f_a(x) = ax(1-x)$ for $0 < a \leq 4$ is known as the logistic family, and is one of the earliest examples where historical behavior is present "abundantly": in [14], it is shown that for an uncountable number of parameters a, Lebesgue almost every orbit in $[0, 1]$ has no asymptotic measure. The example is not persistent, however.

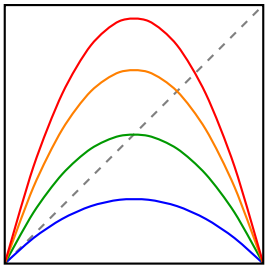


Figure 3.1: The logistic family $f_a(x) = ax(1-x)$.

Example 3.2 (Bowen's eye)**.** Another non-persistent example of historical behavior was given by Bowen (see [15] for the attribution). It can be constructed as the time-one map of a smooth vector field with three fixed points, one of which is repelling, and two saddle points, with two saddle connections, as shown in the figure. The saddles are such that any orbit starting from the interior of the disk bounded by the saddle connections spend arbitrarily large amounts of time close to either saddle point, so that both Dirac masses at the saddle points are limit points of the empirical measures. The "problem" with this example is that the saddle connections are not stable, as shown in the second picture.

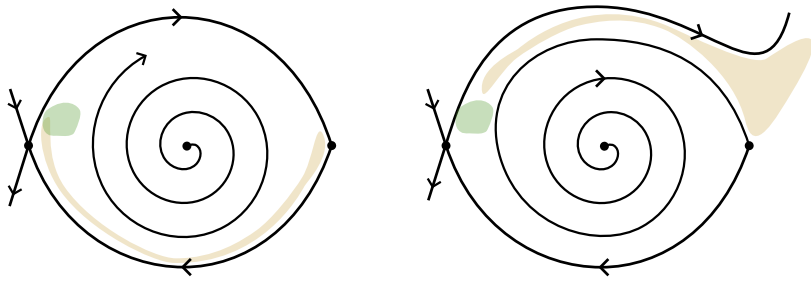


Figure 3.2: Schematic drawing of Bowen's eye, and a perturbation destroying one of its saddle connections.

3.2 Historical behavior for skew products

Let (X, μ) be a Borel probability space, where X is compact and metrizable, and $F\colon X \times I \to X \times I$ a skew product of the form

$$F(x, y) = (g(x), f_x(y))$$

where $g\colon X \to X$ is ergodic and measure-preserving. Suppose there is a homeomorphism $\gamma\colon I \to \overline{\mathbb{R}}$ and a measurable function $p\colon X \to \mathbb{R}$ such that, if we set $t_x(y) = y + p(x)$, we have

$$t_x \circ \gamma = \gamma \circ f_x$$

for all $x \in X$, where t_x is to be interpreted as the translation by $p(x)$.

$$\begin{tikzcd} I \arrow[r, "f_x"] \arrow[d, "\gamma"] & I \arrow[d, "\gamma"] \\ \overline{\mathbb{R}} \arrow[r, "t_x"] & \overline{\mathbb{R}} \end{tikzcd}$$

Since the t_x are translations, there is no loss of generality in assuming $\gamma(0) = -\infty$, $\gamma(1) = +\infty$ and $\gamma\left(\frac{1}{2}\right) = 0$, so we shall adopt this convention for simplicity.

Example 3.3 (Zero Schwarzian)**.** In the example of Bonifant and Milnor from chapter 2, if all the fiber maps f_x satisfy $\mathcal{S}f_x \equiv 0$, then the condition above is satisfied. Indeed, by the last item of proposition 2.3, each f_x is a fractional linear transformation. Therefore, if we define $\gamma(y) = \log y - \log(1-y)$, for $f_x(y) = \frac{a_x y}{1+(a_x-1)y}$ and $p(x) = \log a_x$, we have

$$\begin{aligned} \gamma(f_x(y)) &= \log\left(\frac{a_x y}{1+(a_x-1)y}\right) - \log\left(1 - \frac{a_x y}{1+(a_x-1)y}\right) \\ &= \log\left(\frac{a_x y}{1-y}\right) = \log a_x + \log\frac{y}{1-y} = t_x(\gamma(y)). \end{aligned}$$

In this specific case, we have $f_x'(0) = -f_x'(1) = a_x$. Therefore,

$$\operatorname{Lyap} \mathcal{A}_0 = -\operatorname{Lyap} \mathcal{A}_1 = \int p \, d\mu.$$

By the pointwise ergodic theorem,

$$\frac{\gamma(f_x^n(y)) - \gamma(y)}{n} = \frac{1}{n} \sum_{i=0}^{n-1} p(g^n(x)) \to \int p \, d\mu.$$

Therefore, if $\int p \, d\mu \neq 0$, then either $\gamma(f_x^n(y)) \to \infty$ or $\gamma(f_x^n(y)) \to -\infty$, so all orbits in $X \times (0,1)$ converge to one of the boundaries, which one depending only on the sign of $\operatorname{Lyap} \mathcal{A}_0$.

The interesting case, therefore, is when $\int p \, d\mu = 0$. In the article, the authors managed to show that this case displays historical behavior, but only for the very particular case of $g = kx \pmod 1$ and p being constant on each branch $[i/k, (i+1)/k)$. They conjectured that historical behavior would still be present for the same g but a continuous $p \colon X \to \mathbb{R}$ with zero mean, in general. For the rest of the section, we explain how results already present in the literature, namely weak or almost sure invariance principles, answer their question in the affirmative.

At this point, we recommend that the reader has familiarized themselves with appendix A.4, but we can briefly remind ourselves of some notions defined in there. Let $\xi = (p \circ g^i)_{i \geq 0}$. We can see ξ as a sequence of random variables defined on the probability space (X, μ). Accordingly, we have their partial sums $S_n^\xi = \sum_{i=1}^n \xi_i$ and the sequence of processes $D_n^\xi \colon \Omega \to \mathbb{D}([0,1])$ given by

$$D_n^\xi(\omega)(t) = \frac{S_{\lfloor nt \rfloor}^\xi(\omega)}{\sqrt{n}},$$

as explained in that section. We recall that $W \in \mathcal{M}_1(C([0,1]))$ is the Wiener measure, and that we say ξ satisfies an *weak invariance principle* the processes D_n^ξ converge (weakly) in distribution to σW for some $\sigma > 0$. The theorem below shows that if we assume this condition for g and p, we have historical behavior.

Theorem 3.4. *If the weak invariance principle is valid for the pair (g, p), in the sense $(D_n^\xi)_* \mu \xrightarrow{w} \sigma W$ for $\sigma > 0$ and $\xi = (p \circ g^i)_{i \geq 0}$, then F has historical behavior. More specifically,*

$$\{\delta_-, \delta_+\} \subseteq \{e_n(x,y)\}''_{n \in \mathbb{N}} \tag{3.2}$$

where $\delta_- = \mu \otimes \delta_0$ and $\delta_+ = \mu \otimes \delta_1$ and the limit points are taken in the weak-$*$ topology.

Proof. Let $\delta > 0$ and $s_\delta \colon [0,1] \to \mathbb{R}$ be any continuous function such that $s_\delta(0) = 0$ and $s_\delta(t) > 1 + \delta$ for $t > \delta$. Let $U_\delta = B_\delta(s_\delta)$ be an open ball of radius δ around s_δ in the uniform topology. By theorem A.14, we know $W(U_\delta) > 0$, so by proposition A.6

$$\liminf_{n\to\infty}((D_n^\xi)_*\mu)(U_\delta) \geq \sigma W(U_\delta) > 0.$$

Therefore,

$$S_\delta = \limsup_{n\to\infty}(D_n^\xi)^{-1}(U_\delta) \subseteq X$$

has positive measure. This set is also invariant for g because

$$\begin{aligned}
g^{-1}(S_\delta) &= \limsup_{n\to\infty}\{x \in X \mid D_n^\xi(g(x)) \in U_\delta\} \\
&= \left\{x \in X : \liminf_{n\to\infty}\sup_t \left|\frac{S^\xi_{\lfloor nt\rfloor+1}(x) - p(x)}{\sqrt{n}} - s_\delta(t)\right| < \delta\right\} \\
&= \left\{x \in X : \liminf_{n\to\infty}\sup_t \left|\frac{S^\xi_{\lfloor nt''\rfloor}(x)}{\sqrt{n}} - s_\delta(t)\right| < \delta\right\} \text{ where } t' = t + 1/n \\
&= \left\{x \in X : \liminf_{n\to\infty}\sup_t \left|\frac{S^\xi_{\lfloor nt\rfloor}(x)}{\sqrt{n}} - s_\delta(t)\right| < \delta\right\} = S_\delta.
\end{aligned}$$

Therefore $\mu(S_\delta) = 1$. Now, notice that if $x \in S_\delta$, then there is $n_k^+ \to \infty$ for which for every k, $D_{n_k}^\xi(x) \in U_\delta$. This in turn implies that

$$\#\left\{0 \leq i < n_k^+ \,\middle|\, S_i^\xi(x) > \sqrt{n_k^+}\right\} \geq (1-\delta)n_k^+.$$

Analogously, by repeating the argument with $-s_\delta$, we find $n_k^- \to \infty$ such that for every k,

$$\#\left\{0 \leq i < n_k^- \,\middle|\, S_i^\xi(x) < -\sqrt{n_k^-}\right\} \geq (1-\delta)n_k^-.$$

By considering $E = \bigcap_{i=1}^\infty U_{1/i} \subseteq X$, a set of full measure, and applying a diagonalization argument to the previous two observations, for each $x \in E$ we find $n_k^+, n_k^- \to \infty$ for which

$$\frac{1}{n_k^+}\#\left\{0 \leq i < n_k^+ \,\middle|\, S_i^\xi(x) > \sqrt{n_k^+}\right\} \to 1$$

and

$$\frac{1}{n_k^-}\#\left\{0 \leq i < n_k^- \,\middle|\, S_i^\xi(x) < -\sqrt{n_k^-}\right\} \to 1. \tag{3.3}$$

Now, by ergodicity of g and lemma 1.6, let $S = E \cap \mathcal{B}(\mu)$ be a full measure set where

$$\frac{1}{n}\sum_{i=0}^{n-1}\varphi(g^i(x)) \to \int_X \varphi(z)\, d\mu(z)$$

for all $\varphi \in C(X)$ and $x \in S$. It suffices to show that for all $x \in S$,

$$e_{n_k^-}(x,y) \xrightarrow{w} \delta_- \quad \text{and} \quad e_{n_k^+}(x,y) \xrightarrow{w} \delta_+.$$

We only show the first convergence, since the second is analogous.

Let $\phi \in C(X \times I)$ and $\varepsilon > 0$. Since ϕ is uniformly continuous, there is some k_1 for which

$$\sup\left\{|\phi(x,0) - \phi(x,s)| : x \in X, s \in [0, 1/k_1]\right\} < \varepsilon.$$

Let $D = \{i \mid f_x^{(i)}(y) < 1/k_1\}$, and let N be big enough so that for $k \geq N$,

$$\left| \frac{1}{n_k^-} \sum_{i=0}^{n_k^- - 1} \phi(F^i(x,0)) - \int_X \phi(z,0)\, d\mu(z) \right| < \varepsilon,$$

which is possible since $x \in S$ and $\varphi(x) = \phi(x,0)$ is continuous, and also big enough so that

$$\frac{1}{n_k^-} \#\{0 \leq i < n_k^- \mid i \notin D\} < \varepsilon / \|\phi\|_\infty,$$

which is due to eq. (3.3). Therefore, we can estimate

$$\begin{aligned}
&|e_{n_k^-}(x,y)(\phi) - \delta_-(\phi)| \\
&\quad = \left| \frac{1}{n_k^-} \sum_{i=0}^{n_k^- - 1} \phi(F^i(x,y)) - \int_X \phi(z,0)\, d\mu(z) \right| \\
&\quad \leq \left| \frac{1}{n_k^-} \sum_{i=0}^{n_k^- - 1} \phi(F^i(x,y)) - \frac{1}{n_k^-} \sum_{i=0}^{n_k^- - 1} \phi(F^i(x,0)) \right| + \varepsilon \\
&\quad \leq \frac{1}{n_k^-} \sum_{\substack{0 \leq i < n_k^- \\ i \in D}} \left| \phi(F^i(x,y)) - \phi(F^i(x,0)) \right| + \frac{2}{n_k^-} \sum_{\substack{0 \leq i < n_k^- \\ i \notin D}} \|\phi\|_\infty + \varepsilon \\
&\quad \leq \frac{1}{n_k^-} \sum_{\substack{0 \leq i < n_k^- \\ i \in D}} \left| \phi(F^i(x,y)) - \phi(F^i(x,0)) \right| + 3\varepsilon \\
&\quad \leq \varepsilon + 3\varepsilon = 4\varepsilon.
\end{aligned}$$

This completes the proof. □

Corollary 3.5. *If, in addition to the hypotheses of theorem 3.4, for any compact $K \subseteq (0,1)$ the average visiting time of $f_x^n(y)$ to K tends to zero as $n \to \infty$ for almost every x for some (and therefore, for any) $y \in (0,1)$, that is,*

$$\frac{1}{n} \#\{0 \leq i < n \mid f_x^i(y) \in K\} \to 0,$$

then

$$\{e_n(x,y)\}'_{n \in \mathbb{N}} = [\delta_-, \delta_+] \tag{3.4}$$

for μ-almost every x and all $y \in (0,1)$.

Proof. Let $E \subseteq X$ be a full measure subset of the basin of μ that satisfies both the hypothesis and the result of theorem 3.4. For $x \in E$ and $y \in (0,1)$, let $e_{n_k}(x,y) \xrightarrow{w} \eta$ be any limit point of the empirical measures. First, we show $\operatorname{supp}\eta \subseteq X \times \{0,1\}$. Indeed, let $x_0 \in X$ and $y_0 \in (0,1)$, and let U be a neighborhood of (x_0, y_0) such that $\overline{U} \subseteq X \times (0,1)$. Since $\overline{U}$ is compact and $x \in E$, by hypothesis, $e_{n_k}(x,y)(U) \to 0$, so by Portmanteau

$$\eta(U) \leq \liminf_{k\to\infty} e_{n_k}(x,y)(U) = 0.$$

This shows $(x_0, y_0) \notin \operatorname{supp}\eta$, so $\operatorname{supp}\eta \subseteq X \times \{0,1\}$.

We know η is a limiting measure for x, so it is invariant by F. The fact x is in the basin of μ implies by definition that $(\pi_1)_*\eta = \mu$. This means that $\eta \ll \delta_- + \delta_+$, because for all measurable A,

$$\eta(A) = \eta(A \cap X \times \{0,1\}) \leq \mu(\{x \mid (x,0) \in A\}) + \mu(\{x \mid (x,1) \in A\}) = (\delta_- + \delta_+)(A).$$

Since δ_- and δ_+ are ergodic measures, η must be a convex combination of the two.

We have shown $\mathcal{M}(x,y) \subseteq [\delta_-, \delta_+]$. To conclude eq. (3.2), it suffices to notice that by lemma 1.7, $\mathcal{M}(x,y)$ is connected, and therefore, since by theorem 3.4 it contains both endpoints δ_- and δ_+, it must be the whole segment. □

In the section 3.3, we will show how the weak invariance principle can be derived for a large class of dynamics, including the original $g(x) = kx \pmod 1$.

Infinite ergodic theory

It is possible to show that under the assumption of a weak invariance principle, the extra condition of corollary 3.5 may be met by means of infinite ergodic theory. Let's see some relevant definitions.

Definition 3.6. A map $f\colon X \to X$ on a (potentially infinite) measure space $(X, \mathcal{B}, \mu)$ is called *conservative* if for all measurable sets $E \subseteq X$ with the property that $\{f^{-n}(E)\}_{n\in\mathbb{N}}$,

Remark 4. When the (infinite) measure $\mu \otimes \mathrm{Leb}$ is conservative and ergodic for $(x,y) \mapsto (g(x), y + p(y))$, then the additional hypothesis of corollary 3.5 is satisfied.

todo

Almost sure invariance principle

Alternatively, all the hypotheses of both theorem 3.4 and corollary 3.5 are satisfied when the pair (g,p) satisfies an *almost sure* invariance principle. By that, we mean there exists $\sigma > 0$ and $\delta \in (0, 1/2)$ and new probability space $(\Omega, \mathbb{P})$, along with two sequences of random variables $\zeta = (\zeta_n)_n$ and $\beta = (\beta_n)_n$, where $\zeta_n, \beta_n \colon \Omega \to \mathbb{R}$, such that:

1. ζ and $\xi = (p \circ g^n)_n$ have the same distribution: $\zeta_* \mathbb{P} = \xi_* \mu \in \mathcal{M}_1(\mathbb{R}^{\mathbb{N}})$;
2. β_n are independent and identically distributed as $\mathcal{N}(0, \sigma)$;
3. we can match the partial sums S_n^ζ and S_n^β, in the sense that for $\mathbb{P}$-almost every $\omega \in \Omega$,
$$\frac{S_n^\zeta(\omega) - S_n^\beta(\omega)}{n^\delta} \to 0.$$

In particular, since $(D_n^\beta)_* \mathbb{P} \xrightarrow{w} \sigma W$ by theorem A.15, this condition implies that $(D_n^\xi)_* \mu = (D_n^\zeta)_* \mathbb{P}$ converges to the same limit σW.

Theorem 3.7. *When the pair (g, p) satisfies an almost sure invariance principle, the extra condition of corollary 3.5 is satisfied.*

Proof. Fixing a compact $K \subseteq \mathbb{R}$, item 1 implies that new :)

$$\mu\left(\left\{x \in X \,\middle|\, \frac{1}{n}\sum_{i=0}^{n-1} 1_K(S_i^\xi(x)) \to 0\right\}\right) = \mathbb{P}\left(\left\{\omega \in \Omega \,\middle|\, \frac{1}{n}\sum_{i=0}^{n-1} 1_K(S_i^\zeta(\omega)) \to 0\right\}\right). \tag{3.5}$$

By a change of coordinates using γ, it suffices to show the right-hand side of eq. (3.5) is equal to 1. By choosing $M > 0$ big enough, it is also sufficient to consider $K = [-M, M]$.

Define
$$\Omega_C = \left\{\omega \in \Omega \,\middle|\, \sup_n \frac{|S_n^\zeta(\omega) - S_n^\beta(\omega)|}{n^\delta} \le C\right\}.$$
Notice that by item 3, $\Omega = \bigcup_{C=1}^\infty \Omega_C$.

Fix C, and let $\mathbb{P}_C(A) = \mathbb{P}(A \cap \Omega_C)$. We estimate the rate of decay of $p_n = \mathbb{P}_C(|S_n^\zeta| \le M)$. Let N_C be big enough that $n \ge N_C$ implies $M \le Cn^\delta$. By the triangle inequality, we have $p_n \le \mathbb{P}_C(|S_n^\beta| \le 2Cn^\delta)$ for $n \ge N_C$. Since $S_n^\beta \sim \mathcal{N}(0, n\sigma)$,
$$\mathbb{P}_C(|S_n^\beta| \le 2Cn^\delta) \le \frac{4Cn^\delta}{\sqrt{2\pi n\sigma}} = O(n^{\delta - 1/2}).$$
Therefore, $p_n = O(n^{\delta-1/2})$. Now, let
$$A_n(\omega) = \frac{1}{n}\sum_{i=0}^{n-1} 1_K(S_i^\zeta(\omega)),$$
and let $a_n = \mathbb{E}_{\mathbb{P}_C}(A_n) = \int_{\Omega_C} A_n \, d\mathbb{P}$. By Fubini, $a_n = \frac{1}{n}\sum_{i=0}^{n-1} p_i$, and since $\delta - 1/2 \in (-1/2, 0)$, we also have $a_n = O(n^{\delta-1/2})$.

Let $n_k = 2^k$, $\varepsilon > 0$ and $E_k = \{\omega \in \Omega_C \mid A_{n_k}(\omega) > \varepsilon\}$. By Markov's inequality, $\mathbb{P}(E_k) \le a_{n_k}/\varepsilon$. Since $\delta - 1/2 < 0$, we therefore have $\sum_{k=0}^\infty \mathbb{P}(E_k) < \infty$. Applying Borel-Cantelli,
$$\mathbb{P}(\limsup E_k) = \mathbb{P}(\{\omega \in \Omega_C \mid \limsup_{k\to\infty} A_{n_k}(\omega) > \varepsilon\}) = 0.$$

Since $\varepsilon > 0$ and $C > 0$ are arbitrary, this shows $\lim_{k\to\infty} A_{n_k} = 0$ almost surely. To recover the same limit for the original sequence, we simply notice that $1_K \geq 0$, so

$$A_n \leq A_{n_{k+1}} \cdot \frac{n_{k+1}}{n_k} = 2A_{n_{k+1}}$$

and therefore $A_n \to 0$ almost surely. □

The almost sure invariance principle (ASIP) holds for a large class of dynamics and observables. Here, we highlight the Theorem 2.1 of [16], where it is shown that the ASIP holds for every $\delta > 0$ when g is a nonuniformly expanding map and p is a Hölder continuous function with $\int p\, d\mu = 0$. This includes the original example of Bonifant and Milnor in which $g(x) = kx \pmod 1$.

Question 3.1. Is theorem 3.7 true for the sharp case $|S_n^{\zeta}(\omega) - S_n^{\beta}(\omega)| = o(\sqrt{n})$?

3.3 Deriving the weak invariance principle

The following proof is inspired by the argument at the end of the proof of [17, Theorem 19.1].

Theorem 3.8. *Suppose $p \in L^2(\mu)$ can be decomposed as a sum*

$$p = q + s \circ g - s \tag{3.6}$$

such that $0 < \|q\|_2 < \infty$ and $\zeta = (q \circ g^i)_{i\geq 0}$ is a reversed martingale difference sequence. Then, $\xi = (p \circ g^i)_{i\geq 0}$ is such that $(D_n^{\xi})_*\mu \xrightarrow{w} \sigma W$.

Proof. Notice that

$$S_n^{\xi} = S_n^{\zeta} + s \circ g^n - s,$$

so

$$D_n^{\xi}(x)(t) = \frac{1}{\sqrt{n}} S_{\lfloor nt \rfloor}^{\xi}(x) = D_n^{\zeta}(x)(t) + \frac{1}{\sqrt{n}}\big(s \circ g^{\lfloor nt \rfloor} - s\big)(x).$$

By corollary A.26, $(D_n^{\zeta})_*\mu \xrightarrow{w} \sigma W$, so it suffices to show that for μ-almost every x, the second term in the sum converges uniformly to zero (as a function of t). Given $\varepsilon > 0$, we look at the sets $B_n = \{x : s(g^n(x))^2 \geq n\varepsilon\}$. Since

$$\sum_{n=1}^{\infty} \mu(B_n) = \sum_{n=1}^{\infty} \mu(\{s^2 \geq n\varepsilon\}) = \int \lfloor s^2/\varepsilon \rfloor\, d\mu \leq \frac{1}{\varepsilon}\|s\|_2^2,$$

by Borel-Cantelli, $\mu(\limsup_{n\to\infty} B_n) = 0$. If we take the intersection over k of the sets $\liminf_{n\to\infty} B_n^c$ for $\varepsilon = 1/k$, we get a full-measure set G with the property that for every $x \in G$ and $\varepsilon > 0$, there exists N such that $n \geq N$ implies $\frac{1}{\sqrt{n}}|s(g^n(x))| \leq \varepsilon$. Now, let

$M \geq N$ be big enough so that $n \geq M$ implies $\frac{1}{\sqrt{n}}|s(g^i(x))| \leq \varepsilon$ for all $0 \leq i \leq N$. So for $n \geq M$,

$$\sup_{t\in[0,1]} \frac{1}{\sqrt{n}}\left|s \circ g^{\lfloor nt \rfloor} - s\right|(x) \leq \max_{0\leq i\leq N} \frac{|s \circ g^i(x)|}{\sqrt{n}} + \max_{N\leq i\leq n} \frac{|s \circ g^i(x)|}{\sqrt{i}} \leq 2\varepsilon.$$

□

We will call the decomposition in theorem 3.8 a reverse martingale-coboundary decomposition. In the rest of the section, we derive conditions for it to exist in two different situations: non-invertible, and invertible g.

Non-invertible case

Contrary to the invertible case, when g is not invertible, the sub-σ-algebras $\mathcal{F}_j = g^{-j}(\mathcal{F})$ form a strictly decreasing sequence in general. We will follow [18, Proposition 2.12] for the next theorem.

Theorem 3.9. *Suppose $p \in L^2(\mu)$ and $\mathcal{F}_j = g^{-n}(\mathcal{F})$ are such that*

$$\sum_{n=0}^{\infty} \|\mathbb{E}_\mu(p \mid \mathcal{F}_j)\|_2 < \infty.$$

Then a reverse martingale-coboundary decomposition exists. Moreover, this implies that

$$\int p \, d\mu = \int q \, d\mu = 0$$

and we have the equality

$$\sigma^2 = \|q\|_2^2 = \|p\|_2^2 + 2\sum_{n=1}^{\infty} \int p \cdot p \circ g^n \, d\mu.$$

This gives a way to calculate σ in terms of p. By theorem 3.8, if $\sigma > 0$, then $(D_n^\xi)_*\mu \xrightarrow{w} \sigma W$.

Proof. Denote by $U\colon L^2(\mu) \to L^2(\mu)$ the Koopman operator $U(\varphi) = \varphi \circ g$. Let $L\colon L^2(\mu) \to L^2(\mu)$ be its adjoint:

$$\int L(\varphi) \cdot \psi \, d\mu = \int \varphi \cdot U(\psi) \, d\mu.$$

Since U is an isometry into $U(L^2(\mu)) = L^2(\mathcal{F}_1)$, the restriction $L \mid L^2(\mathcal{F}_1)$ is an isometry as well. Moreover, we have $UL \mid L^2(\mathcal{F}_1) = \mathrm{id} \mid L^2(\mathcal{F}_1)$ (but be careful: $\ker L = L^2(\mathcal{F}_1)^\perp$).

Define

$$s = \sum_{i=1}^{\infty} L^i(\mathbb{E}_\mu(p \mid \mathcal{F}_i)) \quad \text{and} \quad q = \sum_{i=0}^{\infty} L^i(\mathbb{E}_\mu(p \mid \mathcal{F}_i) - \mathbb{E}_\mu(p \mid \mathcal{F}_{i+1})),$$

whose series converge due to the hypothesis and L being an isometry. Rewriting,

$$\begin{aligned}
q &= \mathbb{E}_\mu(p \mid \mathcal{F}_0) + \sum_{i=1}^{\infty} L^i(\mathbb{E}_\mu(p \mid \mathcal{F}_i)) - \sum_{i=0}^{\infty} L^i(\mathbb{E}_\mu(p \mid \mathcal{F}_{i+1})) \\
&= p + \sum_{i=1}^{\infty} L^i(\mathbb{E}_\mu(p \mid \mathcal{F}_i)) - \sum_{i=0}^{\infty} UL(L^i(\mathbb{E}_\mu(p \mid \mathcal{F}_{i+1}))) \\
&= p + \sum_{i=1}^{\infty} L^i(\mathbb{E}_\mu(p \mid \mathcal{F}_i)) - \sum_{i=1}^{\infty} L^i(\mathbb{E}_\mu(p \mid \mathcal{F}_i)) \circ g \\
&= p + s - s \circ g.
\end{aligned}$$

Therefore $p = q + s \circ g - s$. We now check q is a reverse martingale. For that, notice that $L^i(L^2(\mathcal{F}_{i+1})^\perp) = L^2(\mathcal{F}_1)^\perp$, and since $\mathcal{F}_{i+1} \subseteq \mathcal{F}_i$,

$$\int_A \mathbb{E}_\mu(p \mid \mathcal{F}_i) - \mathbb{E}_\mu(p \mid \mathcal{F}_{i+1})\, d\mu = \int_A p\, d\mu - \int_A p\, d\mu = 0,$$

for all $A \in \mathcal{F}_{i+1}$, so $\mathbb{E}_\mu(p \mid \mathcal{F}_i) - \mathbb{E}_\mu(p \mid \mathcal{F}_{i+1}) \in L^2(\mathcal{F}_{i+1})^\perp$. Then, $q \in L^2(\mathcal{F}_1)^\perp$, so $\mathbb{E}_\mu(q \mid \mathcal{F}_1) = 0$ and ξ is an stationary ergodic reverse martingale difference sequence. It's also clear that

$$\int p\, d\mu = \int q + s \circ g - s\, d\mu = \int q\, d\mu = \int \mathbb{E}_\mu(q \mid \mathcal{F}_1)\, d\mu = 0.$$

Also, for any $0 \le i < j$,

$$\int (q \circ f^i)(q \circ g^j)\, d\mu = \int q \cdot (q \circ g^j)\, d\mu = 0$$

because $q \circ g^j \in L^2(\mathcal{F}_j) \subseteq L^2(\mathcal{F}_1)^\perp$. Therefore,

$$\|q\|_2^2 = \frac{1}{n}\sum_{i=0}^{n-1} \|q \circ g^i\|_2^2 = \left\| \frac{1}{\sqrt{n}} \sum_{i=0}^{n-1} q \circ g^i \right\|_2^2.$$

We also have

$$\left\| \frac{1}{\sqrt{n}} \sum_{i=0}^{n-1} p \circ g^i - \frac{1}{\sqrt{n}} \sum_{i=0}^{n-1} q \circ g^i \right\|_2 = \frac{1}{\sqrt{n}} \|s \circ g^n - s\|_2 \le \frac{2\|s\|_2}{\sqrt{n}} \to 0,$$

so

$$\begin{aligned}
\|q\|_2^2 &= \lim_{n\to\infty} \left\| \frac{1}{\sqrt{n}} \sum_{i=0}^{n-1} p \circ g^i \right\|_2^2 \\
&= \lim_{n\to\infty} \frac{1}{n} \Big(\sum_{i=0}^{n-1} \|p \circ g^i\|_2^2 + 2 \sum_{0 \le i < j < n} \int (p \circ g^i)(p \circ g^j)\, d\mu \Big) \\
&= \|p\|_2^2 + 2 \lim_{n\to\infty} \frac{1}{n} \sum_{0 \le i < j < n} \int p \cdot (p \circ g^{j-i})\, d\mu. \qquad (3.7)
\end{aligned}$$

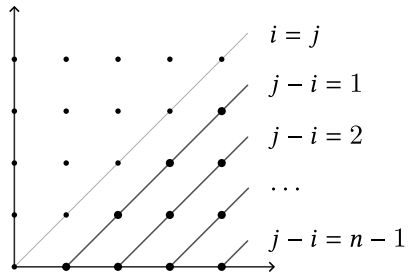


Figure 3.3: Depiction of the identity $\sum_{0 \le i < j < n} f(j-i) = \sum_{i=1}^{n} (n-i) f(i)$.

Now, we have (see fig. 3.3)

$$\sum_{0 \le i < j < n} \int p \cdot (p \circ g^{j-i}) \, d\mu = \sum_{i=1}^{n} (n-i) \int p \cdot (p \circ g^{i}) \, d\mu.$$

Since $p \circ g^i \in L^2(\mathcal{F}_i)$, by Cauchy-Schwarz,

$$\int p \cdot (p \circ g^i) \, d\mu = \int \mathbb{E}_\mu(p \mid \mathcal{F}_i)(p \circ g^i) \, d\mu \le \|p\|_2 \|\mathbb{E}_\mu(p \mid \mathcal{F}_i)\|$$

and therefore

$$\sum_{i=1}^{\infty} \int p \cdot (p \circ g^i) \, d\mu < \infty$$

by the assumption. This implies $i \int p \cdot (p \circ g^i) \, d\mu \to 0$, so the Cesàro average

$$\frac{1}{n} \sum_{i=1}^{n} i \int p \cdot (p \circ g^i) \, d\mu \to 0$$

converges to the same limit. Substituting in eq. (3.7),

$$\|q\|_2^2 = \|p\|_2^2 + 2 \lim_{n \to \infty} \frac{1}{n} \sum_{i=1}^{n} n \int p \cdot (p \circ g^i) \, d\mu = \|p\|_2^2 + 2 \sum_{i=1}^{\infty} \int p \cdot (p \circ g^i) \, d\mu.$$

□

Invertible case

If g is invertible (with measurable inverse), then $\mathcal{F} = g^{-1}(\mathcal{F})$, so there is no hope of applying theorem 3.9. Still, in this case, we may be able to find a smaller σ-algebra $\mathcal{G} \subseteq \mathcal{F}$ for which g^{-1} "forgets" about the past values of p fast enough, but $\mathbb{E}(p \mid g^n(\mathcal{G}))$ is still asymptotically close to p. For the sake of intuition, suppose $X = \{0,1\}^{\mathbb{Z}}$ and g was the shift $g((x_i)) = x_{i+1}$. If we let $\mathcal{G} \subseteq \mathcal{F}$ be the σ-algebra generated by the projections π_j on positive coordinates $j \ge 0$, then for a continuous p, we still expect that $\mathbb{E}_\mu(p \mid g^{-n}(\mathcal{G})) \to 0$ for this algebra (because $\bigcap_{n=1}^{\infty} g^{-n}(\mathcal{G}) = \{\varnothing, X\}$), but we expect $\mathbb{E}_\mu(p \mid g^n(\mathcal{G})) \to p$ (because $\bigcup_{n=1}^{\infty} g^n(\mathcal{G}) = \mathcal{F}$).

Theorem 3.10. *Suppose $p \in L^2(\mu)$, the inverse of g is measurable and there is a smaller σ-algebra $\mathcal{G} \subseteq \mathcal{F}$ for which $\mathcal{G}_j = g^{-j}(\mathcal{G})$ satisfies*

$$\sum_{n=0}^{\infty} \|\mathbb{E}_\mu(p \mid \mathcal{G}_n)\|_2 < \infty \quad \text{and} \quad \sum_{n=0}^{\infty} \|p - \mathbb{E}_\mu(p \mid \mathcal{G}_{-n})\|_2 < \infty.$$

Then a reverse martingale-coboundary decomposition $p = q + s \circ g - s$ for $q \in L^2(\mathcal{G})$ exists and the same conclusions of theorem 3.9 hold.

Proof. Let $U, L \colon L^2(\mu) \to L^2(\mu)$ be adjoint to each other as before:

$$\begin{tikzcd} L^2(\mathcal{G}_i) \arrow[r, bend left, "U"] & L^2(\mathcal{G}_{i+1}). \arrow[l, bend left, "L"] \end{tikzcd}$$

We introduce a new operator $\mathbb{P}_\mu(\phi \mid \mathcal{A}) = \phi - \mathbb{E}_\mu(\phi \mid \mathcal{A})$, which on $L^2(\mu)$ is precisely the orthogonal projection onto $L^2(\mathcal{A})^\perp$. Now, define

$$s^+ = \sum_{i=1}^{\infty} L^i(\mathbb{E}_\mu(p \mid \mathcal{G}_i)) \quad \text{and} \quad q^+ = \sum_{i=0}^{\infty} L^i(\mathbb{E}_\mu(p \mid \mathcal{G}_i) - \mathbb{E}_\mu(p \mid \mathcal{G}_{i+1})),$$

$$s^- = \sum_{i=1}^{\infty} U^i(\mathbb{P}_\mu(p \mid \mathcal{G}_{-i})) \quad \text{and} \quad q^- = \sum_{i=0}^{\infty} U^i(\mathbb{P}_\mu(p \mid \mathcal{G}_{-i+1}) - \mathbb{P}_\mu(p \mid \mathcal{G}_{-i})).$$

All series are absolutely convergent by hypothesis.

□

Remark 5. If p was a coboundary $p = \phi \circ f - \phi$ to begin with, then it is clear that $S_n^\xi = \phi \circ f^n - \phi$, so the iterates of the skew product $F(x, y) = (f(x), y + p(x))$ satisfy

$$F^n(x_0, y_0) = (f^n(x_0), \phi(f^n(x_0)) - \phi(x_0) + y_0) \subseteq \operatorname{graph}(\psi)$$

and are thus contained in the graph of ψ, where $\psi(x) = \phi(x) - \phi(x_0) + y_0$. In particular, F is not ergodic for $\mu \otimes \text{Leb}$.

Corollary 3.11. *Suppose φ is such that there exists a summable sequence $c_n > 0$ satisfying*

$$\int (\psi \circ f^n)\varphi \, d\mu \leq \|\psi\|_2 c_n$$

for all $\psi \in L^2(\mathcal{G})$, where $\mathcal{G}$ is as in theorem 3.9. Then

$$\sum_{n=0}^{\infty} \|\mathbb{E}(\varphi \mid f^{-n}(\mathcal{G}))\|_2 < \infty.$$

Remark 6. The book [18] proves this condition is satisfied for smooth expanding maps, Hölder continuous φ and $\mathcal{G} = \mathcal{F}$, so this includes the $f(x) = kx$ maps. The second condition is trivial when $\mathcal{G} = \mathcal{F}$.

3.4 Recent developments

Conclude with an overview of the recent work of Kiriri, Soma, Vargas et al.

Chapter 4

Partially hyperbolic dynamics

4.1 Partial hyperbolicity

4.2 Invariance Principle

Definition 4.1. A partially hyperbolic diffeomorphism is *accessible* if between any two points in the manifold, there is a path composed of segments tangent to either E^s or E^u.

Definition 4.2. A partially hyperbolic diffeomorphism is *dynamically coherent* if there exists invariant foliations $\mathcal{F}^{cs}$ and $\mathcal{F}^{cu}$ tangent to $E^c \oplus E^s$ and $E^c \oplus E^u$, respectively.

Theorem 4.3. *Let $f\colon M \to M$, where $M = X \times Y$, be a skew product diffeomorphism with an ergodic invariant measure μ. Then, $\lambda^c(\mu) \leq 0$ implies that $\{\mu_x^c\}_{x\in M}$ is invariant by unstable holonomies, meaning that $(h_{x,y}^u)_*\mu_x^c = \mu_y^c$.*

Sketch the proof as in the notes taken on ICTP.
What does it say about the Bonifant-Milnor example.

Appendix A

Measure and probability

A.1 Probability spaces

Products

A fact that is not always covered in measure theory courses and literature is that we can take arbitrary products of measures when they are probability measures. We will sneakily use this fact in some definitions of this section.

Theorem A.1. *Let T be an arbitrary index set, and let $(X_t, \mathcal{B}_t, \mu_t)$ be a probability space. Then, there exists a unique measure μ on the measurable space $(\prod_{t\in T} X_t, \bigotimes_{t\in T} \mathcal{B}_t)$, which is the product of μ_t. That is, for all finite subsets $\Lambda \subseteq T$, considering the canonical projection $\pi_\Lambda \colon \prod_{t\in T} X_t \to \prod_{t\in\Lambda} X_t$, we have $(\pi_\Lambda)_*\mu = \bigotimes_{t\in\Lambda} \mu_t$.*

Proof. See [19, section 3.5]. □

The Kolmogorov Extension Theorem

Let T be an arbitrary index set. Suppose that for every $t \in T$, we have a measurable space $(\Omega_t, \mathcal{B}_t)$. Moreover, suppose we have a more complex specification of how to "patch together" finite collections of these measurable spaces into a measure spaces, that is itself consistent: for every finite subset $\Lambda \subset T$, we specify a measure μ_Λ on $(\Omega_\Lambda := \prod_{t\in\Lambda} \Omega_t, \mathcal{B}_\Lambda := \bigotimes_{t\in\Lambda} \mathcal{B}_t)$, such that if $\Lambda_1 \subset \Lambda_2$, the image of the measure μ_{Λ_2} under the natural projection from Ω_{Λ_2} to Ω_{Λ_1} coincides with μ_{Λ_1}. We call a such a specification $\{\mu_\Lambda\}_\Lambda$ a *projective system* of measures.

From this finitary specification, when is it possible to construct a measure on the whole product $\left(\Omega := \prod_{t\in T} \Omega_t, \mathcal{B} := \bigotimes_{t\in T} \mathcal{B}_t\right)$? Clearly, this is a more general problem than that of constructing an infinite product measure. In fact, this is not always possible

in general, but as we shall see, a topological-flavored condition proves to be sufficient for the existence of the extension.

Definition A.2. Let X be a set. A collection $\mathcal{K} \subseteq 2^X$ of subsets is called a *compact class* if it has the finite intersection property, meaning that for all countable subsets $(K_n)_{n=1}^{\infty} \subseteq \mathcal{K}$, then

$$\left(\forall N, \bigcap_{n=1}^{N} K_n \neq \varnothing\right) \implies \bigcap_{n=1}^{\infty} K_n \neq \varnothing.$$

Definition A.3. Let $(X, \mathcal{B}_X, \mu)$ be a measure space. A compact class $\mathcal{K} \subseteq \mathcal{B}_X$ is called *approximating* if the values of μ on any set can be approximated by those on elements of $\mathcal{K}$, meaning that for any measurable A, and $\varepsilon > 0$, there exists $K \in \mathcal{K}$ such that $K \subseteq A$ and $\mu(A \setminus K) < \varepsilon$.

Theorem A.4 (Kolmogorov Extension Theorem)**.** *Suppose that for every $t \in T$, the measure $\mu_{\{t\}}$ on $\mathcal{B}_{\{t\}}$ possesses an approximating compact class $\mathcal{K}_t \subset \mathcal{B}_{\{t\}}$. Then, there exists a probability measure μ on the measurable space $(\Omega, \mathcal{B})$ such that the image of μ under the natural projection from Ω to Ω_Λ is μ_Λ for each finite set $\Lambda \subset T$.*

Proof. See [19, theorem 7.7.1]. □

Remark 7. Clearly, if each $\mathcal{B}_{\{t\}}$ is the Borel algebra of a topological space whose compact sets satisfy the finite intersection property and $\mu_{\{t\}}$ is inner regular, then it has an approximating compact class given by the compact sets. For instance, this is the case in any Standard Borel space, a family which includes all finite Borel measures in complete separable metric spaces, like $\mathbb{R}$.

A.2 Random variables

If $(X, \mathcal{A}, \mathbb{P})$ is a probability space and $(Y, \mathcal{B})$ is a measurable space, then a *random variable* is simply a measurable function $f\colon X \to Y$. It can be interpreted as some observable quantity of the probability space. If μ is a measure on Y, we may write $f \sim \mu$ to mean that $f_*\mathbb{P} = \mu$, and we say that f has distribution μ.

Definition A.5. We say the collection $\xi = (\xi_i)_{i \in \mathcal{I}} \subseteq X^Y$ of random variables is *independent* if $\xi_*\mathbb{P}$ is a product measure, that is, if

$$\xi_*\mathbb{P} = \bigotimes_{i \in \mathcal{I}} (\xi_i)_*\mathbb{P}.$$

This is equivalent the usual notion that ξ is independent if for every finite subset $J \subseteq \mathcal{I}$ and for every collection of measurable sets $A_j \in \mathcal{B}$, we have

$$\mathbb{P}\left(\bigcap_{j\in J} \xi_j^{-1}(A_j)\right) = \prod_{j\in J} \mathbb{P}\left(\xi_j^{-1}(A_j)\right).$$

A.3 Weak convergence

Proposition A.6 (Portmanteau). *The following are equivalent:*

i. $\mu_n \xrightarrow{w} \mu$;

ii. $\lim_{n\to\infty} \mu_n(\phi) = \mu(\phi)$ *for all Lipschitz functions* ϕ;

iii. $\liminf_{n\to\infty} \mu_n(U) \geq \mu(U)$ *for all open sets* U;

iv. $\limsup_{n\to\infty} \mu_n(F) \leq \mu(F)$ *for all closed sets* F;

v. $\lim_{n\to\infty} \mu_n(A) = \mu(A)$ *for all continuity sets* A *of the measure* μ.

A.4 Brownian motion

The rigorous mathematical foundation of Brownian motion was laid by Norbert Wiener, who constructed Brownian motion as a measure on the space of continuous functions. This construction is why the Brownian motion is often called the Wiener process, and defined as the *Wiener measure*.

We shall denote by $\mathcal{N}(\mu, \nu) \in \mathcal{M}_1(\mathbb{R})$ the probability measure corresponding to the normal distribution with mean μ and variance ν. It can be defined as $\mathcal{N}(\mu, \nu) = \rho\,\mathrm{Leb}$, where ρ is the probability density function

$$\rho(x) = \frac{1}{\sqrt{2\pi\nu}} \exp\left(-\frac{(x-\mu)^2}{2\nu}\right).$$

Definition A.7. Let $\mathcal{C}$ be the product σ-algebra on $C([0,1])$. The Wiener measure W is the unique measure on $(C([0,1]), \mathcal{C})$ defined by the following properties: if we denote by $W_t \colon C([0,1]) \to \mathbb{R}$ the evaluation at time t, seen as a random variable on $(C([0,1]), \mathcal{C}, W)$, then

(i) $W_0 = 0$ almost surely;

(ii) for $t > 0$, $W_t \sim \mathcal{N}(0, t)$;

(iii) if $0 \leq t_1 < t_2 < \cdots < t_n$, then the increments $W_{t_1}, W_{t_2} - W_{t_1}, \ldots, W_{t_n} - W_{t_{n-1}}$ are independent random variables.

The space of càdlàg functions

The space $C([0,1])$ of continuous functions does not let us talk about processes that contain discontinuities, or even about how discontinuous processes can converge in a suitable way to continuous ones. This is important in our setting to deal with the convergence of certain processes induced by discrete dynamics, which consist of step functions, to the Brownian motion, which is continuous almost surely.

The space of *càdlàg* functions, or Skorohod space, is the space of functions whose discontinuities are of the *first kind*, that is, both lateral limits exist, and whose value at the point is equal to the right limit:

$$\mathbb{D}([0,1]) = \left\{ f\colon [0,1] \to \mathbb{R} \,\middle|\, \forall t, \lim_{s\to t^-} x(s) \text{ exists and } \lim_{s\to t^+} x(s) = x(t) \right\}.$$

We want to put a metric on this space with the property that functions are close together if their values are close to each other uniformly, but also allowing small perturbations in the parametrization of time.

Definition A.8 (Skorohod topology)**.** Let Λ be the set of strictly increasing continuous functions from $[0,1]$ to itself. For $f, g \in \mathbb{D}([0,1])$, we define

$$d(f,g) = \inf_{\lambda\in\Lambda} \max\left\{ \|\lambda - \mathrm{id}\|_\infty, \|f - g\circ\lambda\|_\infty \right\}. \tag{A.1}$$

The distance above defines a metric on $\mathbb{D}([0,1])$, whose topology is called the Skorohod topology. Since $d(f,g) \le \|f-g\|_\infty$, the Skorohod topology is coarser than the uniform topology.

Proposition A.9. *The restriction of the Skorohod topology to $C([0,1])$ coincides with the uniform topology. In fact, if $f\colon [0,1] \to \mathbb{R}$ is continuous, then any ball*

$$B_\delta(f) = \{g \in \mathbb{D}([0,1]) \mid \|f-g\|_\infty < \delta\},$$

of the uniform topology contains an open neighborhood of f in the Skorohod topology.

Proof. Since the Skorohod topology is coarser, it suffices to show the second part of the statement. Since f is uniformly continuous, there is $\delta_2 < \delta/2$ such that if $\|\lambda - \mathrm{id}\|_\infty < \delta_2$, then

$$\|f - f\circ\lambda\|_\infty < \delta/2.$$

Let

$$B'(f) = \{g \in \mathbb{D}([0,1]) \mid d(g,f) < \delta_2/2\}.$$

The set $B'(f)$ is a ball of the Skorohod topology, so it suffices to show $B'(f) \subseteq B_\delta(f)$. Let $g \in B'(f)$. Since $\delta_2/2 < \delta_2$, there exists λ such that

$$\max\left\{ \|\lambda - \mathrm{id}\|_\infty, \|g - f\circ\lambda\|_\infty \right\} < \delta_2 < \delta/2.$$

Therefore,

$$\|g - f\|_\infty \le \|g - f \circ \lambda\|_\infty + \|f \circ \lambda - f\|_\infty < \delta.$$

□

Proposition A.10. *If $f_n \to f$ in the Skorohod topology, then $f_n(x) \to f(x)$ for all x in a set E whose complement is at most countable.*

Proof. Since $f \in \mathbb{D}([0,1])$, the set of discontinuities of f is at most countable[1]. Let E be a cocountable set such that $f|_E$ is continuous, and let $x \in E$ and $\varepsilon > 0$. Pick δ so that $|x - y| < \delta$ implies $|f(x) - f(y)| < \varepsilon$, and let N be big enough so that $n \ge N$ implies $d(f, f_n) < \min\{\delta, \varepsilon\}/2$. If $n \ge N$, and if we pick a $\lambda \in \Lambda$ that makes the right side of eq. (A.1) $\min\{\delta, \varepsilon\}/2$-close to the infimum, then

$$\begin{aligned} |f(x) - f_n(x)| &\le |f(x) - f \circ \lambda^{-1}(x)| + |f \circ \lambda^{-1}(x) - f_n(x)| \\ &\le \varepsilon + \|f - f_n \circ \lambda\|_\infty \le 2\varepsilon, \end{aligned}$$

because $|\lambda^{-1}(x) - x| \le \|\lambda - \mathrm{id}\|_\infty < \delta$.

□

The arcsine law

Definition A.11. We shall denote by $\tau^+ \colon \mathbb{D}([0,1]) \to \mathbb{R}$ the observable that returns the proportion of time its function argument is positive, that is,

$$\tau^+(f) = \mathrm{Leb}\{x \in [0,1] : f(x) > 0\} = \int_{[0,1]} 1_{(0,\infty)} \circ f \, d\,\mathrm{Leb}\,.$$

Lemma A.12. *τ^+ is measurable and continuous in a full W-measure set.*

Proof. Since the normal distributions are non-atomic and W_t has normal distribution for every $t \in (0,1]$, we have that

$$W(\{f \in \mathbb{D}([0,1]) : f(t) = 0\}) = 0$$

for all $t \in (0,1]$. Therefore, by Fubini,

$$\mathrm{Leb}(\{t \in [0,1] : f(t) = 0\}) = 0$$

for all f in a set $E \subseteq \mathbb{D}([0,1])$ of full W-measure. It remains to check that τ^+ is continuous at such points $f \in E$. Indeed, 0 is the only discontinuity of the indicator function $1_{(0,\infty)}$,

[1]Proof: let t be a discontinuity point of f, and let U be a neighborhood of t where the value of the function is close to the lateral limits at t. Then, f is continuous at $U \setminus \{t\}$, which is open, and therefore contains a rational point q_t. With some care, this will define an injection from discontinuity points to rationals.

so if $f \in E$ and $f_n \to f$ in the Skorohod space, then $1_{(0,\infty)} \circ f_n \to 1_{(0,\infty)} \circ f$ almost everywhere by proposition A.10. By the dominated convergence theorem,

$$\tau^+(f) = \int_{[0,1]} 1_{(0,\infty)} \circ f \, d\,\mathrm{Leb} = \lim_{n\to\infty} \int_{[0,1]} 1_{(0,\infty)} \circ f_n \, d\,\mathrm{Leb} = \lim_{n\to\infty} \tau^+(f_n).$$

□

Theorem A.13 (Arcsine law). *For $\alpha \in [0,1]$,*

$$(\tau^+)_* W([0,\alpha]) = \frac{2}{\pi} \arcsin \sqrt{\alpha}.$$

Proof. See [17, equation 9.27]. □

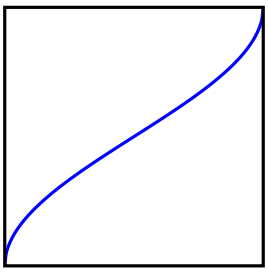


Figure A.1: The value of $\frac{2}{\pi} \arcsin \sqrt{\alpha}$ for $0 \le \alpha \le 1$.

Theorem A.14 (Forgery Theorem, or Support Theorem). *If $f \in C([0,1])$ and $f(0) = 0$, then every neighborhood of f has positive Wiener measure. Note that by proposition A.9, this is true for neighborhoods both in the uniform and in the Skorohod topologies.*

Proof. See, for instance, [20]. This is known as the Stroock-Varadhan Support Theorem or Lévy's Forgery Theorem. □

Weak invariance principles

We will fix a probability space $(\Omega, \mathcal{F}, \mathbb{P})$ for the rest of this section. Given a sequence $\xi = (\xi_0, \xi_1, \dots)$ of random variables defined on Ω, we can define their partial sums $S_n^\xi = \sum_{i=0}^{n-1} \xi_i$ and a sequence of processes $D_n^\xi \colon \Omega \to \mathbb{D}([0,1])$ given by

$$D_n^\xi(\omega)(t) = \frac{S_{\lfloor nt \rfloor}^\xi(\omega)}{\sqrt{n}},$$

obtained by appropriately scaling the partial sums of the random variables up to some iterate n.

Remarkably, for many classes of sequences ξ, their corresponding processes D_n^ξ converge in some way to the Brownian motion — we denote by an *invariance principle* a

result of this form. Perhaps the best known invariance principle is the Donsker's theorem, which is a *weak* invariance principle since it asserts convergence in the weak topology.

Theorem A.15 (Donsker)**.** *If $\xi \subseteq L^2(\mathbb{P})$ is a sequence of i.i.d. variables with mean zero and variance 1, then the measures $(D_n^\xi)_*\mathbb{P}$ converge weakly to the Wiener measure. In other words, for every measurable continuous function $\varphi\colon \mathbb{D}([0,1]) \to \mathbb{R}$,*

$$\int \varphi \circ D_n^\xi \, d\mathbb{P} \to \int \varphi \, dW.$$

The motivation for looking into invariance principles is that they allow transferring certain results from Brownian motions to the sequence of partial sums S_n^ξ. In our case, the result we are interested in is the arcsine law. We know from lemma A.12 that τ^+ is continuous a.e., so weak convergence is enough to transfer its conclusions:

Lemma A.16. *Suppose $\xi \subseteq L^2(\mathbb{P})$ is such that $(D_n^\xi)_*\mathbb{P} \xrightarrow{w} W$. Then the average of the time in which the random walk S_n^ξ is above $M \in \mathbb{R}$ converges to an arcsine distribution, in the sense that*

$$\lim_{n\to\infty} \mathbb{P}\left(\left\{\omega \in \Omega \;\middle|\; \frac{1}{n}\#\{1 \le i \le n \mid S_n^\xi(\omega) > M\} \le \alpha\right\}\right) = \frac{2}{\pi}\arcsin\sqrt{\alpha}.$$

Proof. Writing $\tau_n^+(\omega) = \frac{1}{n}\#\{1 \le i \le n \mid S_n^\xi(\omega) > M\}$, the left-hand side can be rewritten as $\lim_{n\to\infty}(\tau_n^+)_*\mathbb{P}([0,\alpha])$. Set $c_n = M/\sqrt{n}$, and notice that

$$\tau_n^+(\omega) = \frac{1}{n}\#\left\{1 \le i \le n \;\middle|\; \frac{S_i^\xi(\omega)}{\sqrt{n}} - \frac{M}{\sqrt{n}} > 0\right\} = \tau^+(D_n^\xi(\omega) - c_n).$$

Since by theorem A.13 we know $Z = (\tau^+)_*W$ is equivalent to the Lebesgue measure on $[0,1]$, intervals are continuity sets for it. By the weak convergence assumption and continuity of τ^+, we know $(\tau^+ \circ D_n^\xi)_*\mathbb{P} \xrightarrow{w} Z$, and since $c_n \to 0$ (as a sequence in $\mathbb{D}([0,1])$), in fact

$$(\tau^+ \circ (D_n^\xi - c_n))_*\mathbb{P} \xrightarrow{w} Z.$$

By Portmanteau, this means

$$\lim_{n\to\infty}(\tau_n^+)_*\mathbb{P}([0,\alpha]) = \lim_{n\to\infty}(\tau^+ \circ (D_n^\xi - c_n))_*\mathbb{P}([0,\alpha]) = \frac{2}{\pi}\arcsin\sqrt{\alpha}.$$

□

Ergodic Martingale Differences

Donsker's theorem is not the most general result of its kind. In our case, we are interested in a version of the weak invariance principle for ergodic martingale difference sequences. Intuitively, these are the sequences that behave as the time-one steps of a martingale, in the sense that even if the value of some previous steps are known, the expected value of the next step is still zero.

Definition A.17. A sequence $\xi = (\xi_0, \xi_1 \dots) \in L^1(\mathbb{P})^{\mathbb{N}}$ of random variables is said to be a *martingale difference sequence* if for all n,

$$\mathbb{E}_{\mathbb{P}}(\xi_n \mid \sigma(\xi_0, \dots, \xi_{n-1})) = 0.$$

Stationarity and two-sided processes

In the previous definition we considered an one-sided sequence of random variables. As is often the case in ergodic theory, the sequences we shall be dealing with have the additional property of *stationarity*, which is a sort of time-invariance. This property will also allow us to lift one-sided sequences of variables to two-sided ones, that is, sequences with indices in $\mathbb{Z}$, due to the existence of the invertible extensions exposed in section 1.5.

Definition A.18. A sequence $\xi = (\xi_0, \xi_1 \dots)$ of random variables taking values in Y is said to be *stationary* if $\xi_*\mathbb{P}$ is an invariant measure for the left shift $\sigma\colon Y^{\mathbb{N}} \to Y^{\mathbb{N}}$. Similarly, we can define stationarity for two-sided sequences by replacing $\mathbb{N}$ by $\mathbb{Z}$ in the definition.

Definition A.19. A two-sided sequence $\xi \in L^1(\mathbb{P})^{\mathbb{Z}}$ of random variables is said to be a *martingale difference sequence* if for all n,

$$\mathbb{E}_{\mathbb{P}}(\xi_n \mid \sigma(\{\xi_i \mid i < n\})) = 0.$$

Lemma A.20 (Lévy's upward theorem)**.** *Let $(X, \mathcal{B}\mu)$ be a measure space and $\varphi\colon X \to \mathbb{R}$ be a random variable. If $\{\mathcal{B}_n\}_{n\in\mathbb{N}}$ is an increasing sequence of sub-σ-algebras of $\mathcal{B}$ (an upwards filtration), then*

$$\lim_{n\to\infty} \mathbb{E}_{\mu}(\varphi \mid \mathcal{B}_n) = \mathbb{E}_{\mu}\big(\varphi \,\big|\, \sigma(\{\mathcal{B}_n\}_{n\in\mathbb{N}})\big)$$

almost-surely.

Proof. See [21]. □

Lemma A.21. *Every stationary one-sided ergodic martingale difference sequence $(\xi, \mathbb{P}_1)$ can be extended to a two-sided stationary ergodic martingale difference sequence $(\zeta, \mathbb{P}_2)$ with the same marginal distributions on positive indices. That is, if $\pi\colon \mathbb{R}^{\mathbb{Z}} \to \mathbb{R}^{\mathbb{N}}$ is the canonical projection, we have $(\pi \circ \zeta)_*\mathbb{P}_2 = \xi_*\mathbb{P}_1$.*

Proof. Consider the projective system of measures $\{\mu_\Lambda\}_\Lambda$ on $\mathbb{R}^{\mathbb{Z}}$ defined by

$$\mu_{\{z_1, z_2, \dots, z_n\}} = (\pi_{\{0,(z_2-z_1),\dots,(z_n-z_1)\}} \circ \xi)_*\mathbb{P}_1,$$

where for $\Lambda \subseteq \mathbb{N}$, $\pi_\Lambda\colon \mathbb{R}^{\mathbb{N}} \to \mathbb{R}^{\Lambda}$ is the canonical projection. Since $\xi_*\mathbb{P}_1$ is invariant by the left shift σ, this system satisfies the consistency condition of theorem A.4. Let $\mathbb{P}_2 \in \mathcal{M}(\mathbb{R}^{\mathbb{Z}})$

be the extension of the measure given by the theorem, and define $\zeta_n \colon \mathbb{R}^{\mathbb{Z}} \to \mathbb{R}$ simply as $\zeta_n((x_i)_i) = x_n$. By definition, it satisfies $(\pi \circ \zeta)_* \mathbb{P}_2 = \xi_* \mathbb{P}_1$.

Now, we show it is a two-sided martingale difference sequence. By stationarity and the extension property, this two-sided extension satisfies

$$\mathbb{E}_{\mathbb{P}_2}(\zeta_n \mid \sigma(\{\zeta_{n-k}, \dots, \zeta_{n-1}\})) = \mathbb{E}_{\mathbb{P}_1}(\xi_{n+k} \mid \sigma(\{\xi_0, \dots, \xi_{n-1+k}\})) = 0$$

for all k. By letting $k \to \infty$ and applying lemma A.20, this implies $\mathbb{E}_{\mathbb{P}_2}(\zeta_n \mid \sigma(\{\zeta_i \mid i < n\})) = 0$. □

Invariance principle for ergodic sequences

Definition A.22. A sequence $\xi = (\xi_0, \xi_1 \dots)$ of random variables is said to be *ergodic* if $\xi_* \mathbb{P}$ is an ergodic measure for the left shift $\sigma \colon Y^{\mathbb{N}} \to Y^{\mathbb{N}}$.

Example A.23. Let (X, f, μ) be a measure-preserving system with an ergodic measure μ. Any function $\varphi \colon X \to Y$ defines a sequence of random variables

$$\xi = (\varphi \circ f^n)_{n \geq 0},$$

and this sequence can also be seen as a factor map for the extension $\xi \colon (X, f, \mu) \to (Y^{\mathbb{N}}, \sigma, \xi_* \mu)$. By applying lemma 1.12, we see that the sequence ξ is an ergodic sequence.

Theorem A.24. *If $\xi \subseteq L^2(\mathbb{P})$ is a stationary ergodic martingale difference sequence for which $\sigma = \|\xi_1\|_2$ satisfies $0 < \sigma < \infty$, then the measures $(D_n^{\xi})_* \mathbb{P} \xrightarrow{w} \sigma W$.*

Proof. See [17, theorem 18.3]. □

Reverse martingale difference sequences

In ergodic dynamics, the processes we usually care about are the values $\xi = (\varphi \circ f^n)_{n \geq 0}$ of some observable $\varphi \in L^1(\mu)$ along orbits of f. This process is unlikely be a martingale difference sequence unless $\varphi \equiv 0$, because $\varphi \circ f^n$ is measurable for the σ-algebra $f^{-n}(\mathcal{F})$, which is non-increasing due to measurability of f. In other words, we may have $\mathbb{E}_\mu(\varphi \circ f^n \mid \sigma(\varphi, \dots, \varphi \circ f^n)) = \varphi \circ f^n \neq 0$ in general. In the non-invertible case, the algebras $f^{-n}(\mathcal{F})$ form a decreasing sequence in general. This suggests we should look at the processes in the opposite direction, and motivates the following definition.

Definition A.25. A sequence $\xi = (\xi_1, \xi_2 \dots)$ of random variables is said to be a *reverse* martingale difference sequence if for all n,

$$\mathbb{E}_{\mathbb{P}}(\xi_n \mid \sigma(\{\xi_i \mid i > n\})) = 0.$$

In the case of a stationary sequence (as is the case for $\xi = (\varphi \circ f^n)_{n\geq 0}$ and $\mathbb{P} = \mu$ an invariant measure for f), results for martingale difference sequences can be stated dually for reverse martingale difference sequences. The following discussion expands on comments and text present in [17, Appendix M22 and Section 19].

Corollary A.26. *If $\xi \in L^2(\mathbb{P})^{\mathbb{N}}$ is a stationary ergodic reverse martingale difference sequence for which $\sigma = \|\xi_1\|_2 > 0$, then the measures $(D_n^{\xi})_*\mathbb{P} \xrightarrow{w} \sigma W$.*

Proof. First, apply the extension theorem as in lemma A.21 in order to lift ξ to a two-sided sequence $\zeta \in L^2(\mathbb{P}_2)^{\mathbb{Z}}$ given by the coordinates of the probability space $(\mathbb{R}^{\mathbb{Z}}, \mathbb{P}_2)$. Then, invert the sequence: consider $\zeta_i' = \zeta_{-i}$ and the probability $\mathbb{P}_3 = \tau_*\mathbb{P}_2$, where $\tau((x_i)) = (x_{-i})$. Then $(\zeta, \mathbb{P}_3)$ is an ordinary two-sided martingale difference sequence, for which we can apply theorem A.24. Now, the key is noticing that due to commutativity, $\zeta_0' + \cdots + \zeta_n' = \zeta_0 + \cdots + \zeta_{-n}$ is equal to $\zeta_{-n} + \cdots + \zeta_0$, which by stationarity has the same distribution as $\zeta_0 + \cdots + \zeta_n$. The process D_n^{ζ} only depends on such sums, therefore

$$(D_n^{\zeta'})_*\mathbb{P}_3 = (D_n^{\zeta})_*\mathbb{P}_2 = (D_n^{\xi})_*\mathbb{P}$$

for all n, which implies $(D_n^{\xi})_*\mathbb{P} \xrightarrow{w} \sigma W$. □

Appendix B

A categorical lens

B.1 A category

A *category* C consists of a collection of *objects* Ob_C, and for each pair of objects $A, B \in \mathrm{Ob}_C$, a collection of *morphisms* $\mathrm{Hom}_C(A, B)$, also called arrows, endowed with the following properties:

1. given $A, B, C \in \mathrm{Ob}_C$, there exists an operation

$$\circ : \mathrm{Hom}_C(A, B) \times \mathrm{Hom}_C(B, C) \to \mathrm{Hom}_C(A, C)$$

   called categorical composition (which is not necessarily the composition of functions). If $f \in \mathrm{Hom}_C(A, B)$, we may write $f \colon A \to B$, and given $f : A \to B$ and $g : B \to C$, we write $g \circ f := \circ(f, g)$. The composition function must be associative; that is, if $h \in \mathrm{Hom}_C(C, D)$, then $(h \circ g) \circ f = h \circ (g \circ f)$;

2. for every $A \in \mathrm{Ob}_C$, there exists an identity morphism $1_A \in \mathrm{Hom}_C(A, A)$, and for all $f, g \in \mathrm{Hom}_C$ with $\mathrm{dom}(f) = \mathrm{cod}(g) = A$, it holds that $f \circ 1_A = f$ and $1_A \circ g = g$.

B.2 Functors

B.3 Natural transformations

B.4 Limits and colimits

Appendix C

Miscellaneous results

Theorem C.1 (Riesz–Markov–Kakutani)**.** *Let K be a compact space. Then, for every continuous linear functional $L\colon C(K) \to \mathbb{R}$, there exists a unique (possibly signed) Radon measure $\mu \in \mathcal{M}(K)$ such that*

$$L(\varphi) = \int \varphi \, d\mu, \quad \forall \varphi \in C(K).$$

Proof. See [19, Theorem 7.10.4], or most textbooks on functional analysis. □

Corollary C.2. *If X is a compact space, $(\nu_n)_n \in \mathcal{M}(X)^{\mathbb{N}}$ is a sequence of measures and for all $\varphi \in C(X)$, the limit $L(\varphi) = \lim_{n\to\infty} \int \varphi \, d\nu_n$ exists, then $(\nu_n)_n$ converges in the weak-$*$ topology. Moreover, if the ν_n are probabilities, then the weak limit $\nu = \lim_{n\to\infty} \nu_n$ is a probability.*

Proof. By theorem C.1 and definition of weak-$*$ convergence, it suffices to show that L is a continuous linear functional. This, in turn, is a direct application of Banach-Steinhaus to the sequence of linear operators $L_n(\varphi) \mapsto \int \varphi \, d\nu_n$, since the hypotheses imply $C(X)$ is a Banach space. The last observation is simply the fact $\nu(X) = \lim_{n\to\infty} L_n(1) = 1$. □

Proposition C.3. *If a normed vector space E is separable, then the weak-$*$ topology on its dual E' is metrizable. For instance, for a choice $\{\phi_i\}_{i\in\mathbb{N}} \subseteq E$ of a countable dense set,*

$$d(\mu, \nu) = \sum_{i=1}^{\infty} \frac{|\mu(\phi_i) - \nu(\phi_i)|}{2^i},$$

defines a metric.

Proof. See [22, proposition 6.5.1]. □

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